SDMR-STANDARD DATA MONITORING RULES

ICL-COMPRESSOR PERFORMANCES

ICL-COOLING LOOP

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1. Document scope

This SDMR is describing the rules to follow and monitor compressor thermodynamic performances as well as motor cooling efficiency.

Those phenomena are relevant mainly during stabilized operation and therefore power balance and any variable relying on discharge temperatures, such as efficiency or work, will be calculated only for stabilized (thermally speaking) points.

A certain number of calculated values such as volume flows or pressure ratio will be used for monitoring some dynamic condition for example for anti-surge system.

Such variables will be calculated for all available data stabilized or not.

2. General Reference and terminology

2.1. Acronyms and terminology:

PSB: Product structure breakdown. Standard high-level nomenclature (4 to 5 levels) for product defining the block structure used in reliability description and calculation. Incorporate both physical sub-systems but **only** critical instrumentation. This PSB is used to have a uniform product structure description.

IPSB: Instrument product structure breakdown. Standard high-level nomenclature used to describe all standardly available instrument to be used for health monitoring of a product. IPSB uses same structure as PSB at first level but contains only instrumentation description or recalculated variables.

Recalculated variable: Data resulting from mathematical treatment of instruments and convenient for monitoring. Example: Volume flow is usually calculated on orifice combining differential pressure measurement, pressure, temperature, and calibration constants.

Internal/External cooling pattern: In internal cooling pattern the hot cool flow is injected after inlet instrumentation (i.e it is not seen by inlet instrumentation). In external configuration all injection is done before (i.e it is seen by flow and temperature measurement)

Critical instrumentation: Instrument whose failure will lead to unit unavailability either directly (AMB radial displacement sensors) or indirectly through activation of logical trip.

Consistency check / health check: Consistency check will compare two supposed identical or strongly correlated measures or calculated variable. This comparison allows to understand precision of measures associated to a value. For a critical instrument, consistency check allows to check health of a probe. The simplest health check is done in case of redundant instrument by comparing two values and checking for a drift usually tracked by an alarm.

Graph: This term is used generically to define a visualization of a relation Y=f(X1,X2…,t) on a sample of available X1, X2…t corresponding to discrete acquisition or calculation points.

Naming convention: All pressure and temperatures are total ones, t indices for total quantities is omitted in most formulas. Pi,Ti and Po,To are inlet outlet quantities at section inlet. Pext, Tex refers to extraction

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pressure and temperature. For 2 sections cases a numerical index might be added for respectively section 1 and 2.

Project data set: Project dataset is the list of data and software require to either make any-post-treatment specified or compare result of this post-treatment to a reference result or model. Project dataset will include for example flow-metering characteristics, any geometrical data used in post-treatment, normalized or dimensional expected map, reference twin model…. The project dataset is to be constructed prior to post-treatment.

SDMR: Standard Data monitoring rules. Document formalizing the standard "graphs" that are known to be needed for a base preventive or curative monitoring (RCA) of a unit. SDMR are available only for the instrument configurations identified as standard in IPSB.

TWIN-MODEL: A twin model in general sense is a software that calculate a theoretical response function Ytheo=f(X1,X2,….). The software is used on actual samples X1,X2,… and provide as output Ytheo to be compared with measured Y. A simple normalized test curve or tuned CCS can be considered as first level twin-model.

2.2. References:

[ESW0000015] : Reliability for ICL. ICL reliability Product Structure Breakdown and Instrument Product Structure Breakdown.

[SOS 9986830] : "Unit control system standard software requirement". General/functional specification for UCP. This document includes the base data logger functionality requirement but no general specification for historian is present in the document.

In following document XXXX is specific job number trigram associated with standard document name root.

[IPLSHXXXX]: P&ID for the given job. This document describes the final customer tag reference that will be used in all job-related documentation.

[RNTIC060XXX]: UCP I/O list. This document contains typical UCP tag number scale and alarm associated to wired input/output signal on UCP. I/O values are stored in physical units associated to job (bar/psi, °C/°K….). All input data to UCP are normally to be recorded in historian. Output data are likely also required in historian.

[RNTIC064XXX]: UCP communication list (track all exchanges with AMB, VSD…) contains all data list exchanged by UCP including one in Modbus.

Note: Historian must store part of data available UCP communication list in following requirement of SDMR documents for the different subsystem and eventual contact requirement.

RSPIC041XXXX : UNIT CONTROL SYSTEM TECHNICAL SPECIFICATION. This document compiles all technical requirements (general and job specific) for UCP supplier in charge of both hardware and software UCP supply.

Note: By default, data logger requirement are incorporated in UCP, however in case of a complete historian system, supply of system can be either from UCP supplier or from an external source directly followed by BakerHughes.

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2.3. Generalities, generic SDMR target

This document defines the standard rule to monitor a specific sub-system.

SDMR document is intended to be use:

- To define the list and frequency of data acquisition necessary for Historian specification.
- To define the list of standard “graphs” necessary for monitoring of the system.

The target of standard graph is that all the graphs defined in this list shall be pre-created in automated manner (template approach) with possibility for user to de-selected the unused ones.

Objective of monitoring rule is to optimize the use available data (i.e stored in UCP an accessible manually or from a RM&D system) to detect a drift of any parameter symptomatic of degradation of any critical component of unit. Access and interpretation to specialized data stored only in AMB Cabinet or VSD drive (pertu files) is not covered.

This monitoring is intended to be used for either preventive monitoring or curative to support post incident investigations.

Monitoring rules in general consist in either:

- Identifying the variable of interest for a given system called Y. Any measurement associated with trip or alarm defect is a variable of interest but there can be other.
- Comparing Y at two different times. The original value taken after commissioning or maintenance is typically referred as ‘signature’. For example, this is typically done after a maintenance activity to verify the balancing status, reference temperature of bearings….
- For some parameters it is also possible to track evolution of a Y value compared to a calculated value (i.e reference test value for efficiency for example or twin-model value);
- Identify the “drift” of a value over time in the previous comparison.

Drift value can be defined in multiple way. Drift notion is valid for either static/continuous situation or dynamic event.

A static/continuous drift is based on a phenomenon continuously measurable. Efficiency or vibration for example can be monitored and evaluated continuously.

Dynamic drift requires the comparison of discrete “events”; for example, drift of response time of a valve requires detection of all the motion (full opening or closure) of valve and calculation of response time evolution for multiple discrete opening. Dynamic drift identification requires therefor the definition of what is an event and then identification of events in time series.

In the simplest form a drift of a variable can be followed by simply plotting a variable function of time Y=f(t)

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- For dynamic phase (such as start-up) drift is evaluated by superimposing events Y=f(t-t0) Y=f(t-t1),…., Y=f(t-tn).

- For quasi static phase, plotting of Y=f(t) can be used starting from origin.

However, a Y variable usually directly depends on multiple operating or environmental data Y=f(X1,X2,….,t) where X1 are variables for which dependency of Y is obvious from physics and know a-priori.

A drift of Y would require to compare the surface Y=f(X1,X2,….,ti) Y=f(X1,X2,….,ti+1).

If the variation of Y is directly plotted neglecting influence of X1,..,Xn relevant parameters, the variable Y will appear with high noise or apparent drift which are simply coming from change in operating point and not degradation of Y.

Examples: efficiency of compressor varies as a function of flow coefficient and speed. Degradation of performance in time can be monitored only by tracking efficiency change of compressor map and not of efficiency level alone.

Dependence of Y can also be suspected with other physical parameters called secondary parameter. Those secondary parameters are the one to be investigated first to be added to primary X to try establishing an ad-hoc correlation to minimize dispersion at a given time.

In case a large amount of data is available (multiple identical units) it is possible to perform a pure data driven approach to determine the correlations best reducing the variability at a given point in time (i.e leave to a software to determine what are most pertinent x for a given Y)

In general detection of a drift will be done by creating and comparing maps (actual plots) for different period and looking for a drift.

- Y=f1(X1,X2,….t1) période 1 Y=f2(X1,X2,….t2) période 2…… will be traced directly with primary X in standard monitoring rules

- Y=f1(X1,X2,..,x1s,x2s.) période 1 Y=f2(X1,X2,…x1s,x2s.) période 2…… can be hand tested with actual data to see if secondary parameters need to be taken into account

In case the number of X is exceeding 2, comparison of data plot can become difficult, and it is possible also that the available number of samples in a similar range of X is too low to make a simple comparison between measured data set. In such situation it is better to have a comparison with a theoretical model (simple tested map or twin model)

Objective of this document is to formalise the definition of all the parameters to be used for trending or Map plotting.

This document also specifies the intermediate calculated value to be computed in addition to direct measure exploitation.

2.4. Table and color convention:

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Color convention in current document is simply used to simplify reading of document.

Recalculated variable

A recalculated variable is stored in database under a given system indicated in title. It is calculated using a specific mathematical function and using the *variable* stored in the other subsystem. The variable type indicates if recalculated variable is used as X or Y in plots.

In following example, main balance leakage flow in kg/s is calculated by F-orifice function (any orifice leakage calculation), that will use in input a differential pressure and temperature available in compressor system and a pressure in Antisurge sub-system.

Name of calculated variable	Unit	System: Compressor Input variables used for calculation of a calculated variable	Variable type (typical)
Main balance leakage flow (to be stored in section 1)	**kg/s**	**F-orifice (Compressor (*Section1/ balance line differential pressure, Section / temperature*) Antisurge (*Section 2/ Inlet Pressure*))**	**X**

Graph monitoring for Multi variable, consistency check, health check:

A monitoring plot will in general consist in tracing a different time the surfaces Y as a function of all the X. Interpretation of drift will be then to identify if response surface is evolving moving over time.

In case of large number of X, visualization can become complex. Visualization of drift is normally done by:

- Using a theoretical model for variable Y based on inputs Ytheo
- Defining a reference surface response fit (curve fitting of the reference period to create mathematical function)
- Plotting in time either Y-Ytheo and/or Y-Yfit

Of course, it is always possible to directly trace Y as a function of time.

Name (for each section)	Unit	Graph monitoring plot. Combined variable	Variable type
Inlet volume flow	**m^3/s**	**Compressor (Section1/*Inlet Volume Flow*)**	**Y**
Inlet Volume Flow	**m^3/s**	**Process (*Inlet Volume Flow*)**	**X**

Time variable only:

Some variables are only traced as a function of time as no obvious dependency exist. In such a case the table is only containing a Y value.

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Variable name	Unit	Graph monitoring plot. Time	Variable type
Inlet volume flow	**m^3/s**	**Compressor (Section1/*Inlet Volume Flow*)**	**Y**

Specifically, all dynamic events (start-up, valve opening phase…) are defined only as time dependent event. Time dependent events will be typically monitored by plotting different Y in function of time and will be compared to previous event by referring all event to same pseudo time (abscissa shift).

Health monitoring or consistency check

Health monitoring or consistency check is simply the plotting of two variables closely correlated, with same unit and scale, targeting to verify drift of sensor or provide a coherence check.

Such data can be plotted either as a function of time Y and Y' simultaneously or as a 45° plot Y= fct(Y') with identical scale. Deviation in time compared to the 45° line indicates a drift of one variable that can be preliminary indication of probe issue.

For each critical sensor, it is recommended to look for solution to check probe health to anticipate any sensor failure.

Name (for each section)	Unit	Graph monitoring plot. Consistency/health monitoring	Variable
Inlet volume flow	**m^3/s**	**Compressor (Section1/*Inlet Volume Flow*)**	**Y**
Inlet Volume Flow	**m^3/s**	**Process (*Inlet Volume Flow*)**	**Y'**

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3. COMPRESSOR

3.1 Foreword:

Most of calculated variables are making use of thermodynamic calculation associated to an 'external' gas composition that needs to be indicated.

Recalculation of volume or mass flow from diaphragm or estimation of leakages also requires additional geometrical data and use of typical correlations such as Eckert & Schnell or diaphragm/orifice calculation for flow measurement purpose.

Finally, it can be interesting to plot some variable against expected data where expected data are available. This can be the case for efficiency or compressor map plot against actual.

All the expected curves, geometrical data (flow metering), standard gas composition or even digital twin related to the given project shall be stored structurally in project-data structured archive.

All generic

Under compressor systems will be combined all data used to monitor thermodynamic evolution of compressor and partially motor which is interacting.

Modification of performances of compressor is an advanced indicator from increased leakage (wear of labyrinth, or ring failure) or fouling issue on either or both compressor and motor.

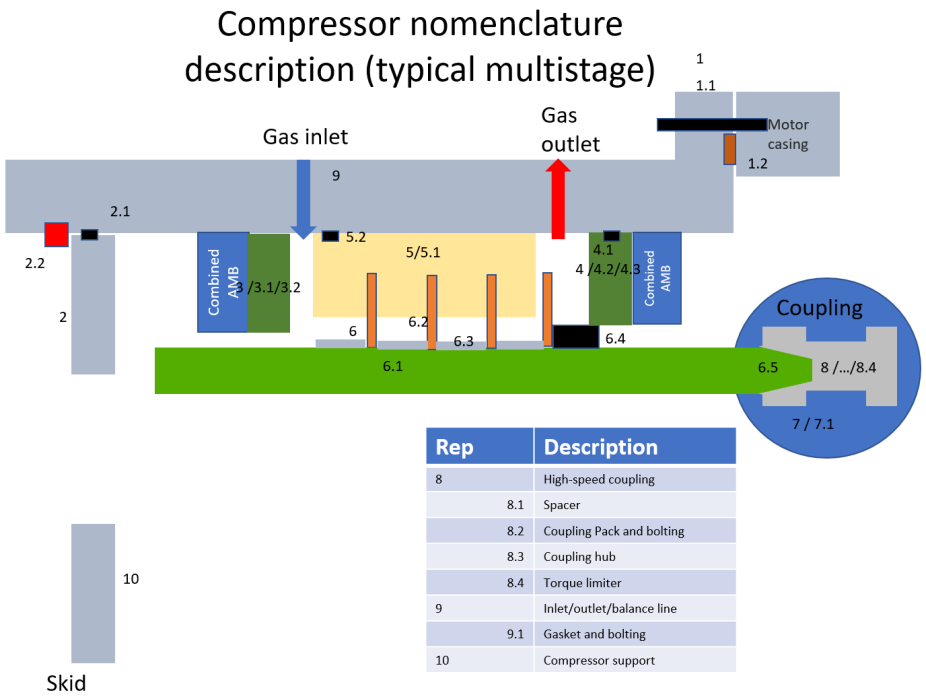
Thermodynamic performances are to be used in quasi static conditions only.

Document for reference

ESW0000015 and associated schematization for ICL compressor PSB and IPSB

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3.2 List of critical instruments (i.e instrument that causes trip if failed) and systems:



Rep	Description
8	High-speed coupling
8.1	Spacer
8.2	Coupling Pack and bolting
8.3	Coupling hub
8.4	Torque limiter
9	Inlet/outlet/balance line
9.1	Gasket and bolting
10	Compressor support

Rep	Description
1	Main flange
1.1	Main flange bolting
1.2	Main flange C-ring
2	Compressor Cover
2.1	Critical gasket
2.2	Shear ring
3	AMB IGV Support
3.1	IGV
3.2	AMB interface
4	Discharge support
4.1	Discharge gasket
4.2	Balance drum assembly
4.3	AMB interface
5	Diaphragm pack
5.1	Labyrinths and bolting
5.2	Critical gasket
6	Rotor
6.1	Shaft
6.2	Impeller
6.3	Sleeves
6.4	Balance drum
6.5	Coupling cone
7	Coupling trap
7.1	Gaskets and bolting

Figure 1: Compressor PSB structure

No critical instrument is present on compressor as per PSB structure.

Monitoring of compressor is done to assess a global decay of thermodynamic performances caused by either compressor issue (leakage, wear, fooling) that can in time lead to more direct operational problems (thrust excess, over-heating, excessive power requirement) …

Additionally, the position of compressor operating points onto compressor map is a critical point for reliability (avoid surge event or spurious landing). In compressor system are produced all data that can be useful to diagnose position of actual operating point onto any stability map.

3.3 List of standard measurement available

PSB and IPSB define under which system are stored the standard critical (trip related) and non-critical (information) sensors. Please consult annex and PSB IPSB files.

For each section, inlet, and outlet flange temperature, which are non-critical measurement, will be stored under compressor system. Pressure measurement which are critical instrument for anti-surge system will be stored under this system.

In case instrument are duplicated, standard monitoring will be done with averaged value.

For a back-to-back configuration with intercooling (cold leakage), availability of measured balance line flow is an interesting feature. A provision for this measurement is often physically present but it is rarely instrumented on site. Different strategies can be proposed to deal with this optional measurement.

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To be noted that most of pressure and differential pressure measurements are critical parameters pertaining to anti-surge system.

3.4 Thermodynamics, gas composition variability:

All post-treatment done in this document require the use of a gas composition and a state equation.

Gas composition will directly influence calculation of volume flow (used in most of XY chart) but also efficiency, head, and power.

Gas composition is almost never part of data continuously measured. At best some sampling will be available from time to time.

As standard post-treatment the gas composition used will be based on data-sheet composition making post-treatment limited to three compositions maximum selected based on molecular weight (minimum average maximum). The average gas composition should normally be aligned with the K coefficient used for standard (programmed in UCP) flow calculation from orifice.

Use of exceptional gas composition (for example N2 clean-up run) is not considered in standard post-treatment.

As a consequence, at least data-sheet selected gas composition should be available in project data-set.

3.5 Single or back-to-back case, internal or external cooling pattern:

Monitoring of thermodynamic performance of an ICL is done by the two following ways:

- Gas and electric power balance
- Efficiency and head monitoring

The power balance provides information on:

- Overall coherence of measurements and indirectly a level of uncertainty of data
- Coherence of estimated gas composition

A brutal variation of power balance is an indication of either gas composition variation or instrument issue.

In such a case, interpretation of any efficiency or head drift should be done carefully.

The way to calculate a proper heat balance and to define and track properly an efficiency value for a system depends on the two following 'configuration' elements:

- Number of sections in compressor. We limit the analysis here to single section and intercooled back-to -back configuration.
- Position of the intercooling (extraction and hot flow return) with respect to available instrumentation (see figure 2)

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3.5.1 Cooling pattern description:

Detail of cooling losses injection point and position with respect to measurement point will influence the interpretation of data.

In the most common case, the standard TCV cooling is associated with internal leakage scheme and the Fan configuration with external leakage scheme.

In the internal scheme (typical standard loop), the power balance is naturally respected. Losses of motor are integrated in section one efficiency number and a common performance assessment of compressor and moment is required calculation motor loss in parallel. “Calculated polytropic efficiency” is poor compared to usual values because it integrates de-facto classically compressor efficiency X motor efficiency (90 to 95%)

For external scheme, compressor performances are independent of motor performances. Compressor measured efficiency is and motor efficiency.

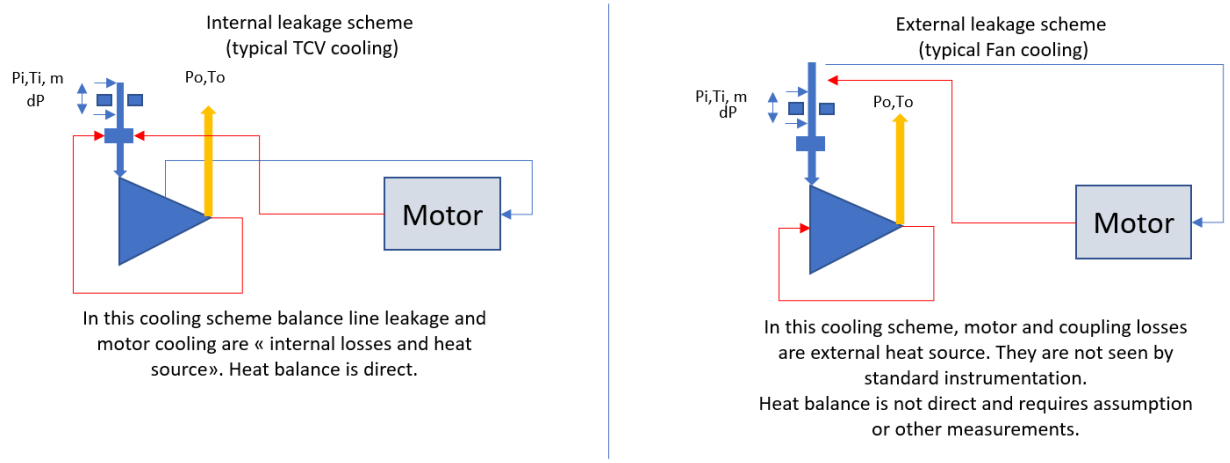


Figure 2 : Consequences of cooling flow re-injection point onto power balance writing.

For internal configuration scheme (typical of standard cooling line), a coherence check is done with motor power.

An eventual deterioration of motor performances or cooling valve will be integrated to section 1 efficiency or will be evidenced in the cooling loop monitoring.

For external scheme, typical of FAN use, the extraction and injection point are before measurement.

The external scheme allows to have section 1 efficiency level comparable to conventional compressor and the cooling losses can be calculated.

3.5.2 In line and back-to-back, power balance:

In this chapter we describe the main element for power balance of either a single section or a back-to-back compressor with intercooling and how to calculate the section gas power. Calculation is valid for either a internal or external leakage configuration (see figure 2 for definition).

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Single section, the gas power, section head and polytropic head are simply calculated with inlet and outlet quantities. Use the formulation of equation 1 for a general calculation valid for single or double section back-to-back with or without intercooling. Depending on leakage scheme configuration and the position of measurement point, the heat and power balance will be interpreted differently.

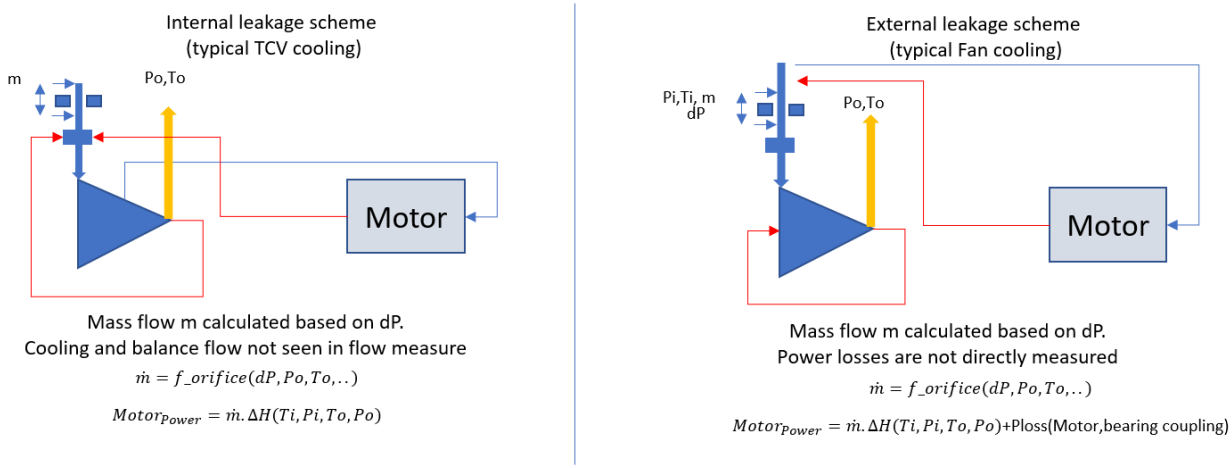


Figure 3 : Head calculation and power balance of a single section compressor

3.5.3 Impeller 1 optional flow measurement and process flow for ICL:

The historical flow monitoring systems for ICL are the one represented on fig.3.

One technical issue of this monitoring scheme specifically for internal leakage configuration is that the actual volume flow in stage 1, which is driving stability of compressor, differs heavily from the measure process due to large cooling flow. Anti-surge monitoring and logic is therefore not very precise.

The other drawback of configuration can be simply the cost od the specific flow monitoring device.

In certain configuration of ICL, a second type of device making a pseudo-flow measurement has been implemented. This device is a simple differential pressure measurement taken between inlet flange and impeller eye.

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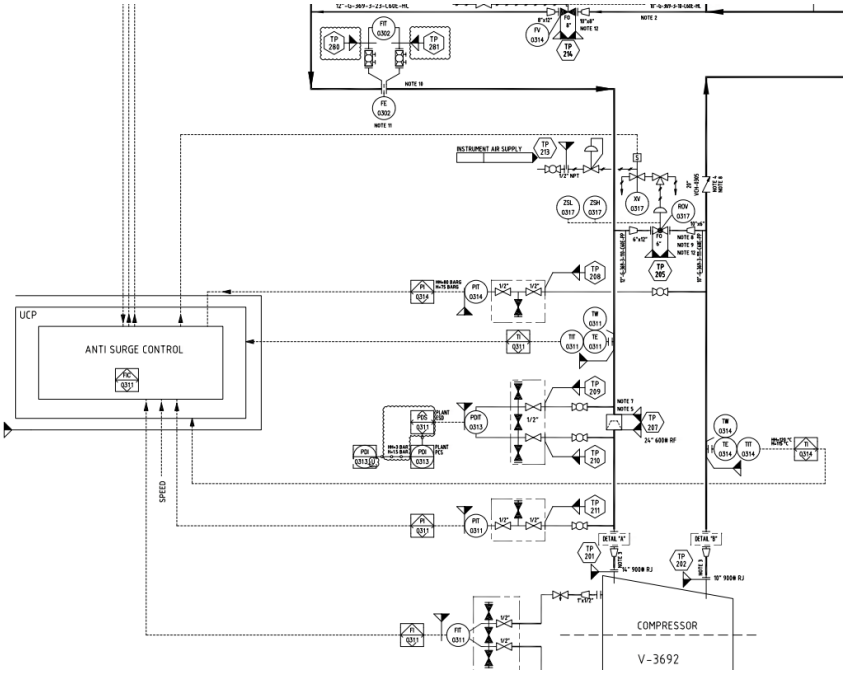


Figure 4:Orifice flow measurement with impeller 1 flow measurement installed for anti-surge.

This device operates like an orifice and based on a calibration of a K factor, allows to have a 'direct' measurement of impeller inlet volume flow. This impeller 1 direct flow measurement (fig.4) is used only for anti-surge control purpose and does not appear in flange volume flow based on classical flow element in BH scope or customer flow measurement.

Impeller volume flow is noted Qv' in this document. In standard rule this Qv' is a completely calculated value based on equation of following chapter.

In case the optional pseudo flow differential pressure is available the Qv' (calculated) will be compared to Qv'm value based on measurement.

3.5.4 General mixing equation for section 1:

For different reasons, it is interesting to define by calculation the different intermediate mixing station at section depending on mixing of the balance drum of main balance line and the cooling flow.

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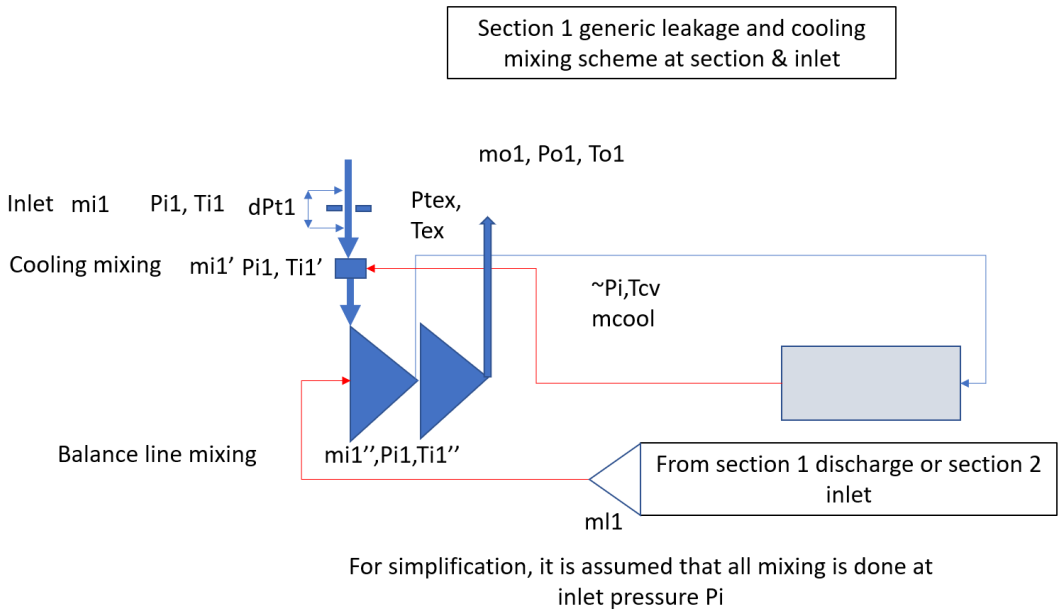


Figure 5: General mixing definition for cooling and balancing of section 1.

With simplified assumption that all mixings are done at inlet pressure it comes the following equations to define mass flow and mixed enthalpy after respectively cooling flow mixing and balance line leakage mixing:

$$mi1' = mi1 + mcool \text{ (a)}$$

$$H(pi, Ti1') = \frac{mi1(H(Pi1, Ti1) + mcoolH(Pi1, TCV)}{mi1 + mcool} \ (b)$$

$$mi1'' = mi1 + mcool + ml1 \text{ (c)}$$

$$H(pi, Ti1'') = \frac{mi1(H(Pi1, Ti1) + mcoolH(Pi1, TCV) + ml1.H(Pl1, Tl1)}{mi1 + mcool + ml1} \ (d)$$

Equations 1 a,b,c and d allow to calculate intermediate mixing quantities (temperature) and associated mass and volume flow.

With Pl1,Til1 being the thermodynamic condition in balance line associated to balance leak flow ml1 i.e Po1,To1 for a single section or Pi2,Ti2 for a back to back.

The balance leak flow ml1 is often not directly measure but can be calculated.

The equation above are written assuming that average cooling temperature at motor discharge and cooling flow mcool are known.

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3.6 List of calculated variables for compressor

Calculated variables typically incorporate all variables required for performance assessment such as volume flow, actual or polytropic head, recalculated power etc.…

Internal values are calculated only using information derived from Baker Hughes supplied instrumentation.

External values are using customer Process information eventually computed via other instrumentation or re elaborated from Baker Hughes Probes.

Internal and external information might not be always simultaneously available.

3.6.1 Single section:

For a single section configuration, the following data will be calculated directly based onto inlet and outlet pipe instrumentation as per Figure 1.

Name of calculated variable	Unit	System: Compressor Input variables used for calculation of a calculated variable	Variable type (typical)
Inlet volume flow	m^3/s	thermos equation + Orifice function orifice(Compressor (Section1/Inlet temperature) Antisurge (Section1/Inlet Differential pressure , Section 1/Inlet Pressure))	X
Specific volume flow (phi)	Adim	Analytic function (8.Qv/(pi.D^3.w) (D impeller standard diameter) Compressor (Section1/Inlet Volume Flow) Compressor AMB (Speed sensor)	X
Section Inlet mass flow	Kg/s	Thermos equation for density X volume flow Compressor (Section1/Inlet Volume Flow, Section1/Inlet temperature) Anti surge (Section1/Inlet Pressure)	X
Section polytropic head	Kj/kg	Thermo equations Anti-surge (Section1/Inlet Pressure, Section1/Outlet Pressure) Compressor (Inlet temperature, Oulet temperature)	Y
Section Specific polytropic thead (psi)	Adim	Compressor (Section1/ polytropic head) Compressor AMB (Speed sensor) (Nota Psi = Section pol head/(Speed^2))	
Section head	Kj/kg	Thermo equation:Anti-surge (Section1/Inlet Pressure, Section 1/Outlet Pressure) Compressor (Section1/Inlet temperature, Section1/Oulet temperature)	Y

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Section specific head (tau)	**Adim**	**Compressor (Section1/ Actual head) Compressor AMB (Speed sensor) (Nota Tau = Section head/(Speed^2))**	**Y**
Section Polytropic efficiency	**Adim**	**Thermo equation Compressor (Section1/ polytropic head internal, Section1/ Actual head)**	**Y**
Section specific speed (u/a0)	**adim**	**Thermo equation for a0 Compressor AMB (Speed sensor)** **Compressor (Section1Inlet temperature) Antisurge (Section1/Inlet Pressure)**	**X**
Section power	**kW**	**See general formulation for gas power with leakage correction for btb**	**Y or X**
Section Pressure ratio	**adim**	**Pressure ratio of section**	**Y**
Specific mixed volume flow (phi')	**Adim**	**Use quantities Equation 1 a and b. Internal leak case only** **See 2.4.3 for inlet variable**	**X**
Measured mixed volume flow (phi'm)	**Adim (optional)**	**Impeller 1(Compressor (Section1/Inlet temperature)** **Antisurge (Section1/impeller1 Differential pressure , Section 1/Inlet Pressure))**	**X**
Mixed U/a0'	**Adim**	**Use quantities Equation 1 a & b. Internal leak case only** **See 2.4.3 for inlet variable to calculate ti'**	**X**
Eta'	**Adim**	**Polytropic efficiency based on T' as per 2.4.3**	**Y**
Psi'	**adim**	**Polytropic head based on T' as per 2.4.3**	**Y**
Discharge flow coefficient phi''	**Adim**	**Compressor (Section 1Inlet mass flow,Section1/inlet temperature) Anti-surge (Section1/Outlet pressure) Compressor AMB (Speed sensor)**	**X**
Discharge U/a0	**Adim**	**Compressor (Section1/outlet temperature) Anti-surge (Section1/Outlet pressure)**	**X**
Motor relative efficiency	**Adim**	**Cooling loop(Motor efficiency)*Gaspower section 1/Shaftpower**	

Tableau 1: General thermodynamic monitoring parameters

3.6.2 Back-to-back configuration:

In case of a two sections compressor, recalculated data for section2 will be identical to section1. Compared to section 1 calculation, main differences are:

- So far no direct pseudo flow measurement is used on section 2 first impeller
- Leakage calculation for a back to back to recalculate mixed first impeller condition is different

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For a typical back-to-back with intercooling, the most typical cooling configuration is standard TCV because of the relatively large number of stage and possible extraction points and internal cooling scheme.

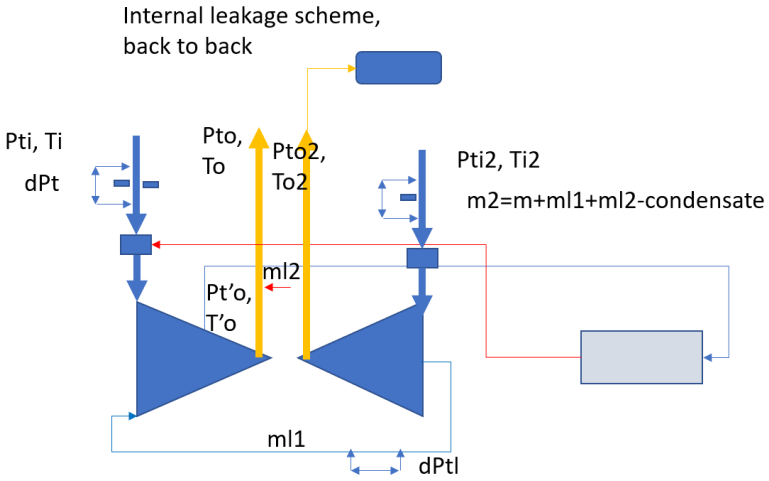
For this configuration it is required to have additional flow measurement device onto the main cold balance line (ml1) to close the power balance equation. The provision for this device is normally available but most of time not connected to transmitter. In case a direct measurement is not available, a calculation can be used instead based on assumption of a given clearance.

With a direct measurement of ml1 (cold balance leakage), it is possible to obtain a 'calculated' ml2 provided (as a process data) that condensate mass flow at scrubber is measured or calculated.

In absence of any specific process extraction injection at interstage except for eventual condensate extraction balance measurement, it comes:

mi2=mi1+ml1+ml2-mlcondensate

Equation 2: Mass balance for a back-to-back



$$(\dot{m} + ml1).\Delta H(Pt'out, Tt'out, Pti, Tti) + (\dot{m2} - ml1).\Delta H(Pto2, Tto2, Pti2, Tti2) = Pmoteur$$

The data indicated with ' are calculate data prior to mixing.
Full closure of balance requires measure of internal leakage flow ml1 and condensate flow calculation.
With those data one can calculate also the value of ml2

Figure 6: Back-to-back typical available measurement

After writing equation in function of intermediate mixed out condition it comes in presence of the two leakages, ml1 and ml2, formulation for gas power:

$$Gas_{power}section\ 1 = m.\big(H(Po,To) - H(Pti,Ti)\big) - ml2(H(Po2,To2) - H(Po,To)) + ml1(H(Po,To) - H(Pl2,Tl2))$$ (a)

$$Gas_{power}section\ 2 = m2.\big(H(Po2,To2) - H(P2i,T2i)\big) - ml1(H(Po2,To2) - H(Pti2,Ti2))$$ (b)

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*The notation Pl2 Tl 2 refers to the balance leakage origin point. For a single section Pl2=Po,Tl2= T0 and the last term vanishes.

Equation 3: Gas power calculation for a back-to-back compressor or in line compressor. General formulation with leakage incorporation and internal leakage scheme

In first equation, term 1 is the usual gas power definition. The second term implies that the leakage ml2 do not contribute to section & actual gas power. It simply reheats the exit; The final term implies that the leakage flow ml1 is compressed by section 1.

In second equation, the first term is the classical gas power while the second term implies that the leakage ml1 does not pass-through section 2.

Where m an m2 are respectively measured mas flow at flange (internal cooling)

Even at compressor level, the sum of total gas power comes as a function of ml1 and ml2.

Precise Power balance of a back-to-back can only been done with a direct measure of ml1.

ml2 can be indirectly computed via a mass balance et section 2 inlet and usually comes.

If ml1 is not measured, the power balance will be with poor precision, and it will not be possible and double check of indication of cool loss will not be possible.

Polytropic efficiency:

For simplicity, in case of a back to back, the polytropic efficiency for each section will be calculated in standard way using Po,To,Pi,Ti and Po2,To2,Pi2,Ti2.

Leakage surface variation will be used to monitor an eventual leakage clearance increase.

Name of calculated variable	Unit	System : Compressor Input variables used for calculation of a calculated variable	Variable type (typical)
Balance leakage flow 1 (to be stored in section 1)	**kg/s**	**F-orice (Compressor (Section1/ balance line differential pressure, Tl) Antisurge (Pl)) *Pl,Tl are pressure and temperature at leakage point origine, either discharge of section & or inlet of section 2**	**X**
Balance leakage flow 2	**Kg/s**	**F-laby (Compressor (Tl) Antisurge (Pl)) External info: Gas type, orifices size *Pl,Tl: see above**	

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Discharge flow leak 1 or 2 (to be stored in section 2, store under Discharg flow leak 2 when based on balance leakage flow 2)	**Kg/s**	**From** m2=m1+ml1+ml2-mlcondensate calculate **Compressor (Section2 Mass flow,balance leakage mass flow, Condensate mass flow) Or F-laby (Compressor (Section 2 discharge temperature) VI Antisurge (Section 1 discharge,Section 2 Discharge Pressure))** **External info: Gas type,**	**X**
Condensate mass flow (to be stored in section 2)	**Kg/s**	**From thermo equation make Flash calculation Compressor (Section 1/ Mass flow, Section 2/ inlet temperature) Antisurge (Section 2/Inlet Pressure) External info: Gas type,**	**X**
Balance leak surface (to be stored in section 1)	**m2**	**For case with direct measure of ml1 and ml2, calculate leak surface from measure mass flow and laby correlation F-laby (Compressor (Section 2 inlet temperature) Antisurge (Section 1 inlet pressure,Section 2Inlet Pressure))**	**Y**
Inter section leak surface (to be stored in section 1)	**m2**	**For case with direct measure of ml1 and ml2, calculate leak surface from measured mass flow and laby correlation F-laby (Compressor (Section 2 temperature) Antisurge (Section 1 inlet pressure,Section 2Inlet Pressure))**	**Y**

Tableau 2 Specific calculated variable for leakage calculation. Single section or back-to-back compressor

The leak parameter will be calculated using a classical formula (typical Eckert & schnell) or other software using unitary leakage surface.

Leakage=S.F(P2,P1,T1,Gas, tooth number & nominal clearance.)

The orifice calculation (or equation corresponding with the as is metering device) will be used for mass flow calculation from actual measurement.

3.6.3 Note on Motor efficiency impact, cooling power definition:

For a standard cooling option, return of motor hot gas after flange measurement on section 1 has a direct impact on the measured flange to flange calculated efficiency. Effect of motor presence onto section 1 polytropic efficiency will depend onto the ratio CoolingPower/Section 1 gas power (for single section gas power).

The above ration is called motor relative efficiency. For a single section compressor, it is equal to motor efficiency.

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It will be calculated based on motor efficiency and section 1 gas power. Motor efficiency calculation will be treated in detail in chapter 3 for TCV option where it has the largest impact.

Cooling power is defined as the sum of cooling work+Motor+coupling+AMB losses. The base equation for cooling power comes:

$$Coolpower = mcool.\Delta(H(Pex, Tex - Pi, Ti'') + Motorloss + Couplingpowerloss + AMBloss$$

Equation 4: Cooling Power definition

Where:

- Motor and AMB losses are sum of both winding and electrical (all type) losses whereas coupling is winding
- Pex,Tex are thermodynamic conditions at extraction point.
- Pi and Ti" are mixed pressure and temperature (i.e after cooling and balance line flow injection)

Once Cool Power calculated, motor efficiency=Coolpower/Shaft-Power is calculated

3.7 IPSB structure for compressor (see ESW00015)

The final IPSB structure for compressor contains a relatively large number of data, mainly recalculated variables that are classical ones for compressor efficiency follow-up or surge analysis. The motor efficiency is another important variable specific to ICL especially important with standard cooling option.

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em Id	System description	Level 2 id	Description	Level3 i	Description	Note
		1	Section 1			
				1	Inlet Temperature	
				2	Discharge temperature	
				3	Balance line differential pressure	
				3	Inlet Volume Flow	
				4	Phi	
				5	Inlet Mass Flow	
				6	Polytropic head	
				7	Psi	
				8	Actual head	
				9	Tau	
				10	Polytropic efficiency	
				11	U/a0	
				12	Discharge flow coefficient	
				13	u/a0 discharge	
				14	phi'	
				15	Eta'	
				16	u/a0'	
				17	Section Power	
				18	Motor relative efficiency	
				19	Balance leakage flow 1	Si mesure directe
				20	Balance leak surface	
				21	Balance leakage flow 2	Si calcul
				22	phi''	
I	Compressor			23	Eta''	
				24	u/a0''	
		2	Section 2			
				1	Inlet Temperature	
				2	Discharge temperature	
				4	Inlet Volume Flow	
				5	Phi	
				6	Inlet Mass Flow	
				7	Polytropic head	
				8	Psi	
				9	Actual head	
				10	Tau	
				11	Polytropic efficiency	
				12	U/a0	
				13	Discharge flow coefficient	
				14	phi-leak	
				15	Section Power	
				16	Condensate mass flow	
				17	inter-section mass flow	
				18	inter-section leak surface	
				19	Discharge flow leak 1	Si mesure directe pour la section 1
				20	Discharge leak surface	Si mesure directe pour la section 1
				21	Discharge flow leak 2	
		3	Overall			
				1	Gas power	
				2	Motor relative efficiency	
		1	Radial-AMB-DE			
				1	Stator temperature sensors(*)	
				2	Radial position sensors	
		2	Radial-AMB-NDE			
					Stator temperature	

PSB | IPSB | Feuil1

Accessibilité : consultez nos recommandations

3.8 Consistency check:

Some values can be calculated from different sources from BH instrumentation (example Power) or customer (i.e volume flow or mass flow based on customer station information).

When two supposed identical values are present, plot the 45°graph and check for eventual drift of data in time.

The consistency check will be used to diagnosis an eventual drift of instruments and help interpretation of other data.

All data that are only available from customer system will be generically stored under the system Process.

Name (for each section)	Unit	Graph plot	Variable
Inlet volume flow	**m^3/s**	**Compressor (Section 1/ Inlet Volume Internal)**	**Y**

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Inlet Volume Flow	**m^3/s**	**Process (Inlet Volume Flow)**	**X**

When a direct flow measurement is present on impeller 1, comparison with the measured and calculated flow coefficient (or volume flow) is giving an indication on eventual drift of instrument or clogging.

Name (for each section)	Unit	Graph plot	Variable
Phi'	**Adim**	**Compressor (Section 1/ phi')**	**Y**
Phi'm	**Adim**	**Compressor (Section 1/ phi'm)**	**X**

Name (for each section)	Unit	Graph plot	Variable
Inlet mass flow	**kg/s**	**Compressor (Section1/Inlet Mass Flow)**	**Y**
Inlet Mass Flow	**kg/s**	**Process (Inlet Mass Flow)**	**X**

The power balance for the unit will be traced. Note that for a back-to-back configuration, power balance requires availability of a balance line measured flow. In case this quantity is not available, theoretical leakage surface will be used. Direct measurement of balance line is therefore recommended feature for a back-to-back.

Name (for each section)	Unit	Graph plot	Variable
Gas Power	**kW**	**Compressor (Overall/Gas Power)**	**Y**
Motor Power	**kW**	**VSDS (Power electronics/Shaft Power)**	**X**

<u>For internal measurement configuration (standard TCV):</u>

Gas Power and Motor Power should match within few % for a single section compressor if a proper flow metering system is used. In back-to-back configuration, expected precision is less.

The power balance for internal configuration allows to quantify precision of measurement and interpret performance drift.

For external measurement (Fan or semi closed loop) a delta should exist in theory representing the cooling power. See coherence check for those two cases.

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3.9 Single section performance monitoring rules:

Performances of a compressor is typically associated to a compressor map function with Volume Flow/rotating speed, Rotating Speed / Inlet Mach (U/a0), Gas Composition as input.

Output are Polytropic head, Pressure ratio, efficiency, and gas power.

Depending on the configuration of cooling, efficiency will be influenced by motor cooling losses (TCV with internal leakage) or not (Fan with external configuration or semi-closed loop.

The normalized graph of efficiency versus phi will help to reduce dispersion. This graph should be used first to monitor efficiency drift.

Polytropic efficiency	Unit	Graph plot	Variable type
Polytropic efficiency internal	Adim	Compressor (Section1/ Polytopic efficiency)	Y
Phi	Adim	Compressor (Section1/phi)	X1
U/a0	adim	Compressor (u/a0)	X2
Motor relative efficiency	kW	Compressor (Motor relative efficiency)	X3

Psi and Tau Map will then be used for complementary analysis of efficiency drift

Polytropic head	Unit	Graph plot	Variable type
Psi	Adim	Compressor (Section 1/Psi)	Y
Phi	Adim	I Compressor (phi)	X1
Normalized speed	adim	Compressor AMB (Speed sensor)	X2
Motor relative efficiency	kW	Compressor (Motor relative efficiency)	X3

Work coefficient	Unit	Graph plot	Variable type
Tau	Adim	Compressor (Section1/Psi)	Y
Phi	Adim	Compressor (phi)	X1
Normalized speed	adim	Compressor AMB (Speed sensor)	X2
Motor relative efficiency	kW	Compressor (Motor relative efficiency)	X3

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*For the configuration with semi-closed loop or external configuration, the term X3 is not influent on compressor efficiency. X3 parameter dependance is to be removes.

For the standard cooling loop, it is interesting to plot an internal compressor efficiency which is getting read of the motor cooling flow effect in term of temperature (Eta'). Calculation of ' quantities is done as per & 2.4.3 equation 1 2 and 3 and requires evaluation of motor cooling flow, motor losses and eventually leakage with either available instrument or best estimation possible.

Polytropic efficiency internal	Unit	Graph plot	Variable type
Eta'	Adim	Compressor (Section 1/Polytopic efficiency')	Y
Phi'	Adim	Compressor (phi')	X1
U/ao'	adim	compressor (u/ao')	X2

Polytropic head internal	Unit	Graph plot	Variable type
Psi'	Adim	Compressor (Section 1/Psi')	Y
Phi'	Adim	Compressor (phi')	X1
U/ao'	adim	compressor (u/ao')	X2

Optional tracing for back-to-back:

In case a direct measurement of balance leakage is done (recommended for back-to-back), it is recommended to trace performance evolution also in function of phi'' which is the actual flow after complete mixing.

This tracing can help to interpret if balance leak alone (piston wear) is sufficient to explain completely an eventual performance variation.

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3.10 Two sections back-to-back case:

3.10.1 Consistency check:

For a configuration with presence of balance leakage measurement, the evolution of the “measured” leak surface will be plotted.

In absence of measurement, the surfaces used for leak calculation will be theoretical only or as measured (constant).

Variable	Unit	Graph lot	Variable type
Balance leak surface	**m^2**	**Compressor (Section1/balance leak surface)**	**Y**
Intermediate leak surface	**m^2**	**Compressor (Section2/balance leak surface)**	**Y**

The surface calculated above should be of same order of magnitude as the surface in the drawing and can be monitored directly in time.

In case of drift of leakage value on the above charts, wear of the labyrinth is a likely cause and a corresponding decay in compressor efficiency should be also visible.

In case of erratic variation of surface variation or order of magnitude of surface not corresponding to drawing, measurement quality or unsteadiness of data is likely cause.

3.10.2 Section 1 monitoring:

Section 1 monitoring will be done according to single section monitoring scheme.

For a back-to-back configuration, the efficiency measured tends to be higher than for a one section since balance leakage is cooled (cold to cold leak) and the discharge leakage (hot to hot) is in principle not visible. However, in case of second section inlet and outlet having temperature highly different from section 1 respective inlet and outlet, a high noise can appear in performance plot.

Plotting of section 1 efficiency after correction of leakage and cooling flow is recommended.

3.10.3 Section 2 monitoring:

Monitoring of performances of second section of a back-to-back compressor is usually simpler. Unless for a shift in actual flow coefficient, the efficiency is not affected by leakages (which go from section 2 to section 1) and section 2 is not affected normally by cooling scheme. Section 2 monitoring is therefore identical to a classical nonintegrated compressor.

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The second section efficiency on the contrary is behaving like a basic single compressor section and can be monitored with simple phi,psi eta mapping

Internal efficiency	Unit	Graph plot	Variable type
Polytropic efficiency	Adim	Compressor (Section2/Polytopic efficiency)	Y
Phi	Adim	Compressor (Section 2/ phi)	X1
u/a0	Adim	Compressor (section 2/U/a0)	X2

Psi and Tau Map will then be used for complementary analysis of efficiency drift

Psi (section 2)	Unit	Graph plot	Variable type
Psi	Adim	Compressor (Section2/Psi)	Y
Phi	Adim	Compressor (Section 2/ phi)	X1
u/a0	Adim	Compressor (section 2/U/a0)	X2

Tau (section 2)	Unit	Graph plot	Variable type
Tau	Adim	Compressor (Section2/Tau)	Y
Phi	Adim	Compressor (Section 2/ phi)	X1
u/a0	Adim	Compressor (section 2/U/a0)	X2

For surge monitoring, the same variable will be traced according to discharge flow coefficient and all discharge flow coefficient corrected from leakage.

A drift in the parameters of efficiency for section 2 will indicate either external leakage increase (shift in phi axis), fouling (shift in Y axis) or internal leakage increase (shift in both X and Y axis)

For surge monitoring and specifically in case of balance leakage measurement, it is interesting to plot the corrected inlet flow coefficient (after extraction of leakage) and the three equivalent plot using discharge flow coefficient (with assumed inlet flow, inletml1 flow and inlet -ml1-ml2)

Specifically in case of wearing of labyrinth, the actual flow in section 2 vs flange flow will tend to reduce and position in surge map may become inadequate.

Psi (section 2)	Unit	Graph plot	Variable type

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Psi	**Adim**	**Compressor (Section2/Psi)**	**Y**
Phi-leak	**Adim**	**Compressor (Section 2/ discharge flow coefficient)**	**X1**
u/a0'	**adim**	**Compressor (Section 2/ U/A0')**	**X2**

Psi (section 2)	Unit	Graph plot	Variable type
Psi	**Adim**	**Compressor (Section2/Psi)**	**Y**
Discharge flow	**Adim**	**Compressor (Section 2/ phi-leak1)**	**X1**
u/a0'	**adim**	**Compressor (Section 2/ U/A0')**	**X2**

Psi (section 2)	Unit	Graph plot	Variable type
Psi	**Adim**	**Compressor (Section2/Psi)**	**Y**
Discharge flow-leak1	**Adim**	**Compressor (Section 2/ phi-leak2)**	**X1**
u/a0'	**adim**	**Compressor (Section 2/ U/A0')**	**X2**

3.11 Recording and interpretation of data:

Operating condition for record

Stabilized points only will be used for any thermodynamic calculation

Sampling

1 point per minute up to 1 point per hour for stabilized operation

Recommended scale for plotting Y,X

Min, Max scale will be used for all variables.

Record time archiving

Continuous monitoring. A yearly download and year to year comparison is recommended.

Example of record (will be added later)

TBD

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Control to be done:

Upon modification of performances in map or motor losses.

Check with customer possible gas composition drift.

Use consistency check to verify possible instrument drift (or recalibration). Double check with site activities

If no issue is found, modification of performances 'out of noise' can be sign of fouling of motor (internal configuration), compressor wear or gasket issue (after rebundling). Double check with thrust drift in bearing group and in cooling line diagnosis.

Recommended action according control conclusion:

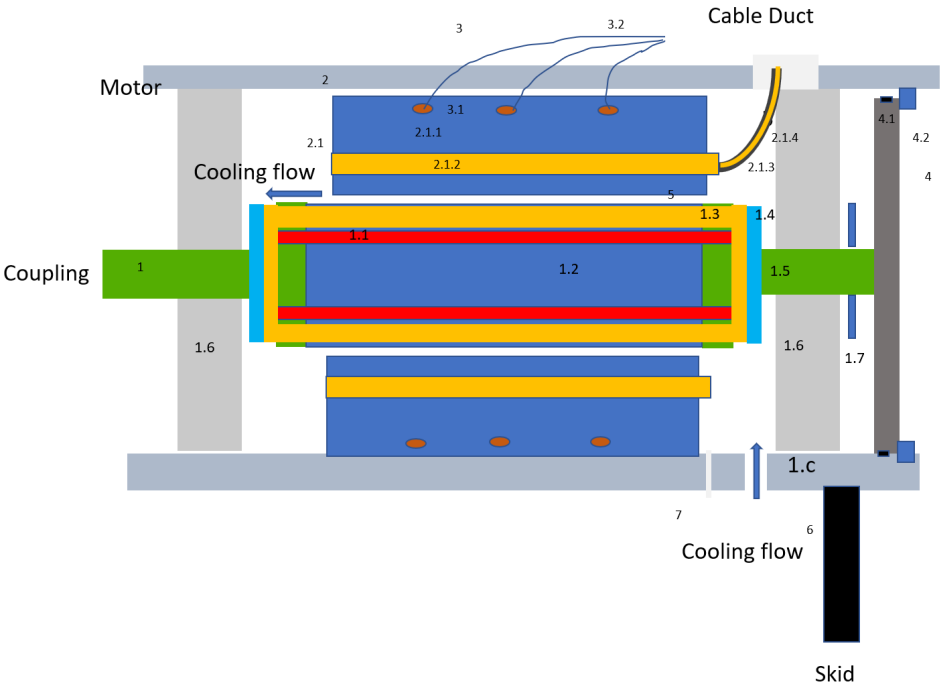
If drift confirmed to be significant, most likely action on next maintenance will incorporate o-ring and labyrinth check and change, possible bundle cleaning and eventual motor cleaning for suspicion of motor fouling.

Assess vibration level, planned maintenance schedule spare part for final decision.

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4 Motor

4.1 Critical system and instrumentation



Rep	Description
1	Rotor
1.1	Tie-rods
1.2	Lamination packing
1.3	Squirrel cage
1.4	Flux damper
1.5	End-heads
1.6	Bearing set (thrust optional)
1.7	Fan (optional)
2	Motor Casing
2.1	Stator structure
2.1.1	Stator Lamination pack
2.1.2	Power coils
2.1.3	Power cable
2.1.4	Power cable insulation and protection
3	Stator Instrumentation
3.1	Temperature probes
3.2	Probes wiring and insulation
4	Motor casing cover
4.1	Critical gasket
4.2	Shear ring
5	Air Gap
6	Motor legs
7	Drains

The only critical instrumentation stored under motor system will be the stator temperature winding.

4.2 List of calculated variables:

Some calculated variables resulting or used in post-treatment of cooling performances are common to all cooling systems and will be stored under motor system:

Name of calculated variable	Unit	System: Motor Input variables used for calculation of a calculated variable	Variable type
Motor efficiency	**Adim**	**Cooling loop standard/Fan/ Seimiclosed (cooling power1 or 2 3) / VSDS(Power electronic/Shaft Power)**	**X**
Swirl ratio	**Adim**	**See equation 5**	**X**
Shaft Torque	**N.m or adim**	**VSDS (Power electronic/Shaft Power)/AMBMotor(Motor speed)**	**X**

The swirl ratio is defined as coefficient for a rotating disk in the form:

$$\text{Swirl ratio } \left(\frac{\rho}{\rho ref}\right)^{1-\varepsilon}\left(\frac{w}{wref}\right)^{3-\varepsilon}\left(\frac{Rey}{Reyref}\right)^{\varepsilon}$$

Équation 5:Swirl ratio

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The coefficient is normally taken at 0.2. A tuning, best fit or more elaborated correlation using also cooling volume or mass flow can be used or tested with sufficient data available

4.3 Coherence check:

For multiple reason, the speed information is critical to VSDS, motor and compressor. The three different values can be monitored in time.

Name (for each section)	Unit	Graph plot	Variable
Rotating speed	RPM	Compressor AMB (speed sensor)	Y
Rotating speed	Rpm	Motor AMB (speed sensor)	Y
Rotating speed	Rpm	VSDS (speed set point)	Y

4.4 Monitoring Rules

The winding temperature coherence and monitoring check are described in cooling system

Name of calculated variable	Unit	Graph	Variable type
Motor efficiency	Adim	Cooling loop standard/Fan/ Semiclosed (cooling power1 or 2 3) / VSDS(Power electronic/Shaft Power)	Y
Swirl ratio	Adim	See equation 5	X1
Shaft Torque	N.m or adim	VSDS (Power electronic/Shaft Power)/Motor-AMB(Speed sensor)	X2
Shaft-Power	kW or adim	VSDS (Power electronic/Shaft Power)	X3

To avoid multiplication of variable, the cooling power to be used will be the most reliable available that is cooling power 1 in case return line is instrumented. In absence of direct monitoring, cooling 2 will be used.

Because the cooling power also includes the fan power or the first stage compression power of cooling flow, the losses are not only coming from friction (swirl ratio) and torque / shaft power but will also vary (at second order) depending on cooling flow compression power. This will add secondary variables that are:

- For TCV mainly: Compressor inlet temperature, inlet pressure
- For Fan or semi closed loop: Rotating speed

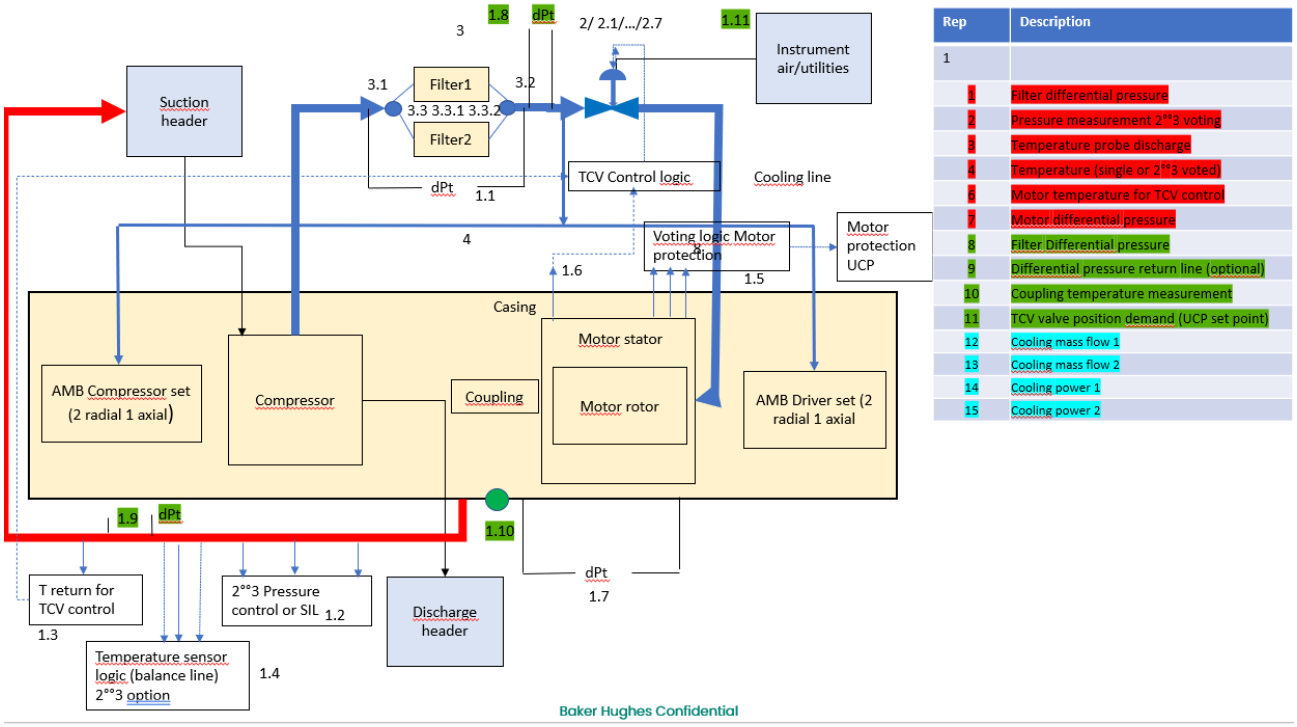
Analytical models start to be helpful to help motor efficiency drift interpretation if 5 variables are present. Interest to add compressor inlet pressure and temperature will depend on the variability of those parameters.

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5 Cooling loop standard with TCV

5.1 List of critical instrumentation:

Critical instrumentation from PSB (VII Standard Cooling loop) is describe in the following:



Rep	Description
1	
1	Filter differential pressure
2	Pressure measurement 2°°3 voting
3	Temperature probe discharge
4	Temperature (single or 2°°3 voted)
6	Motor temperature for TCV control
7	Motor differential pressure
8	Filter Differential pressure
9	Differential pressure return line (optional)
10	Coupling temperature measurement
11	TCV valve position demand (UCP set point)
12	Cooling mass flow 1
13	Cooling mass flow 2
14	Cooling power 1
15	Cooling power 2

Figure 7: Critical instrumentation list for standard cooling loop (please refer to ESW000015)

Critical systems to be monitored are:

In addition to critical instrumentation for which standard health monitoring is to be done, critical system to be monitored are:

- motor performances (efficiency) from typical fouling. For this configuration, motor performances are partially interacting with and captured onto section 1 performances

- TCV functionalities

5.2 List of standard measurement available:

In addition to the critical instrumentation list, differential pressure measurement station is usually available on return line and on inlet line. However, those station are not always equipped with transmitters and cooling flow might only be provided in case of requirement.

A temperature monitoring is present onto coupling chamber and used for monitoring coupling temperature.

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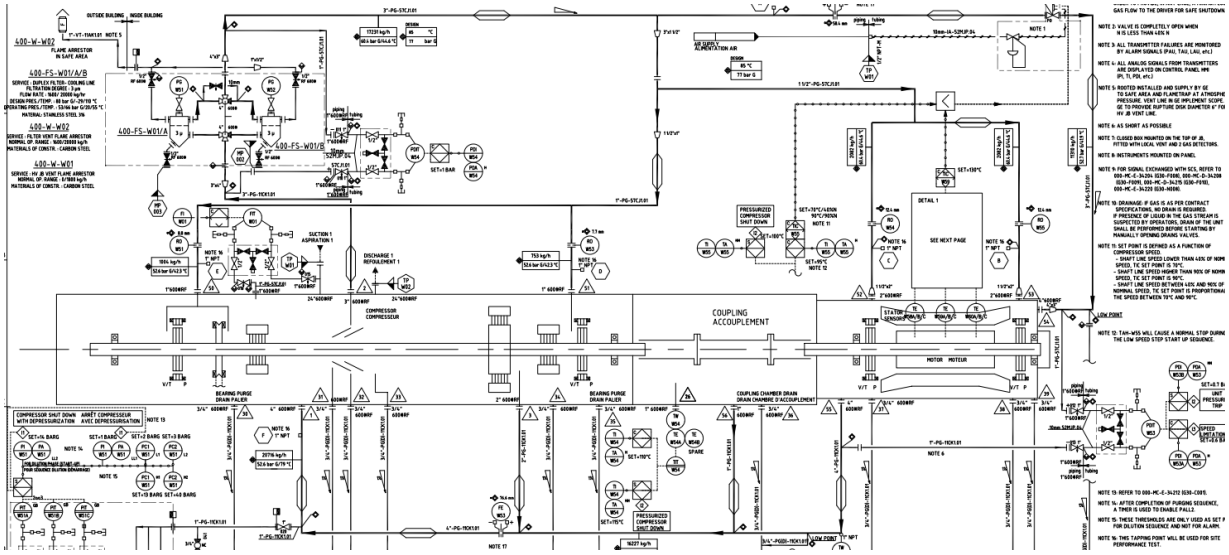


Figure 8: Typical P&ID for a Cooling standard TCV depicting provision for flow element.

5.3 List of calculated variables:

Cooling flows and cooling powers

The most important recalculated variables are the cooling mass flows.

Cooling mass flow can be measured using principally return line differential pressure, temperature, and pressure (identical to inlet) and give cooling flow 1.

A second indirect way to measure cooling flow is using motor differential pressure as a “cheap” monitoring device which is giving cooling mass flow 2.

In principle, cooling flow 1 will be based on simple diaphragm flow calculation and gives more reliable information of mass flow compared to cooling flow 2 for which a speed dependency exists and relationship with flow is more theoretical. However cooling flow 1 is rarely available on recent units contrary to cooling flow 2 for which instrumentation is always present.

Relationship assumed for cooling mass flow is taken from RNT000138 and is in the form:

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4. Motor Pressure Losses across the coil radial clearance (ALL ICL)

$$\Delta P_{motor_air_gap} = K_{motor_air_gap} * \frac{(0.8 * Q_{motor})^2}{\rho_{motor_inlet}}$$

$\Delta P_{motor_air_gap}$	: Pressure drop	[bar abs]
$K_{motor_air_gap}$	: EM pressure losses coefficient – see table below	[ξ motor air gap / Area ^{2} Motor air gap]
Q_{motor}	: Electric motor total mass flow	[kg/s]
ρ_{motor_inlet}	: Density at the motor inlet	[kg/m^3]

Applicable Motor	$K_{motor_air_gap}$ -
MGV0 IH(L1) 135	0.20813
MGV0 Not define yet	
MGV1 IH(L1) 112	0.75310
MGV1 IH(L1) 091	0.75310
MGV2 IH(L1) 085	1.80850
MGV2 IH(L1) 075	1.80850
MGV2 IH(L1) 067	1.80850
MGV3 IH(L1) 067	4.37003
MGV3 IH(L1) 053	4.37003

Table 8 : EM PRESSURE LOSSES COEFFICIENT (ALL ICL)

Figure 9 Typical assumed relationship between cooling flow and differential pressure.

The equation above is of the form:

$$dP = K.0.64\ Qm.Qv = K.0.64 * S^4.\rho.Vm^2 \ (1)$$

The equation chosen is of similar form of the one of a fix orifice.

The eventual dependency of the coefficient with motor length, swirl ratio(Vm/w.r), Reynolds… are neglected.

Calculation of ro (density at motor inlet) will be based on pi Ti +(To-Ti)/ n with n being stage number which is another approximation necessary here.

Also, in previous equation, the cooling flow is not integrating some of the flow going to bearing (motor an compressor DE) necessary for full power balance.

<u>Cooling power and cooling flow relationship:</u>

A cooling flow is assumed available from previous chapter. We want now to calculate the cooling power defined by following equation

$$Coolpower = mcool.(H(Pex, Tex) - H(Pi, Ti")) + Motorloss + Couplingpowerloss + AMBpartial\ loss$$

The first term of the equation is the power loss linked to the compression of the mass flow used in cooling loop. The other terms are motor losses and AMB losses (motor bearing and DE compressor bearing) which end-up in the cooling line. NDE AMB loss will end-up onto compressor efficiency measurement.

Where

Pi,Tim are mixed condition after cooling flow return and mixing with inlet flow (Pi,Ti)

Pex,Tex are condition at extraction pont

Tim and Tex are not measured but can be calculated

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Once a cooling mass flow has been estimated, the mixed-out temperature Tim can be calculated by enthalpy balance considering no pressure loss i.e Pim =Pi and considering that return flow temperature is measured on return line

$$H(pi, Ti1'') = \frac{mi1(H(Pi1, Ti1) + mcoolH(Pi1, TCV) + ml1.H(Pl1, Tl1)}{mi1 + mcool + ml1}$$

$$Coolpower = mcool.(H(Pex, Tex) - H(pim, ti")) + (mcool).(H(pI, Tcv) - H(pex, tex)$$

$$Coolpower = mleak.(H(Pi, Tcv) - H(pi, ti"))$$

Finally, once a cooling mass flow is selected, cooling power is calculated using following typical equation:

$$Coolpower = mcool.\left(H(Pi, Tcv) - \frac{mi1(H(Pi1, Ti1) + mcoolH(Pi1, TCV) + ml1.H(Pl1, Tl1)}{mi1 + mcool + ml1}\right)$$

Equation 6: Relationship between cooling flow and cooling power

The cooling power will therefore be calculated based on the best available estimation between mcool and ml1 (measured quantity if available in cooling flow 1, or calculated quantity in cooling flow 2)

Name of calculated variable	Unit	System: Cooling loop standard Input variables used for calculation of a calculated variable	Variable type
Measured Cooling flow 1	**Kg/s**	**Cooling loop standard (Differential pressure return line, Return line temperature,Pressure measurement)**	**Y or X**
Calculated Cooling flow 2	**Kg/s**	**Cooling loop standard (Motor differential pressure, Return line temperature,Pressure measurement) Motor AMB (Speed sensor)**	**Y or X**
Coolpower 1	**kW**	**Cooling loop standard (cooling flow 1 Return line temperature,Pressure measurement) Compressor (Ti, mass flow)**	**Y**
Coolpower 2	**kW**	**Cooling loop standard (cooling flow 1 Return line temperature,Pressure measurement) Compressor (Section 1/Inlet temperature, Section1/mass flow)**	**Y**

As a normal practice, the cooling flow and cooling power that will be used will be first based onto measured quantities when available.

5.4 Consistency check / health monitoring:

Health monitoring for Motor differential pressure

In the configuration where two cooling flows can be calculated, the two will be compared. The coherence between the two calculated values will be verified.

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Variable	Unit	Graph plot	Variable type
Cooling flow 2	Kg/s	Cooling loop standard (cooling flow 2)	Y
Cooling flow 1	Kg/s	Cooling loop standard (cooling flow 1)	X1

And same for the cooling power

Variable	Unit	Graph plot	Variable type
Cooling power 1	kW	Cooling loop standard (cooling flow 2)	Y
Cooling power2	kW	Cooling loop standard (cooling flow 1)	X1

Health monitoring for Return Line temperature:

Health monitoring for temperature will be done simply by comparing casing temperature with return vane temperature. Casing and coupling temperature are assumed to follow one another

Variable	Unit	Graph plot	Variable type
TCV control	°C	Cooling loop standard (Return line temperature)	Y
Coupling temperature monitoring	°C	Cooling loop standard (Coupling temperature)	X1

Health monitoring for motor temperature:

Motor temperature is expected to be more closely following the coupling and return line temperature.

Variable	Unit	Graph plot	Variable type
Motor temperature voted	°C	Motor (Stator/winding temperature)	Y
Coupling temperature monitoring	°C	Cooling loop standard (Coupling temperature)	X1
TCV control temperature	°C	Cooling loop standard (Return line temperature)	X2

Health monitoring for Filter differential pressure and motor differential pressure:

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For short period of time i.e with filter not having time to foul, filter differential pressure and motor differential pressure are both function of mass flow^2

Variable	Unit	Graph plot	Variable type
Filter differential pressure	**Pa**	**Cooling loop standard (Motor differential pressure)**	**Y**
Motor differential pressure	**Pa**	**Cooling loop standard (Filter differential pressure)**	**X**

It is expected that the two variables are proportional. There will be a progressive variation in time coming primarily from filter clogging. An erratic variation or a sudden drift can be tracked and should be either symptom of a normal filter change or from a potential reading issue.

Health monitoring for 'dilution pressure':

A specific pressure measurement is present on cooling loop to ensure that pressure is not going below pre-defined pressure use to maintain pressurisation status (avoid air inlet).

Variable	Unit	Graph plot	Variable type
Return line pressure	**Pa**	**Cooling loop standard (Return line Pressure)**	**Y**
Section 1 inlet pressure	**Pa**	**Cooling loop standard (Inlet Pressure)**	**X**

5.5 Cooling System Monitoring rules:

Motor fouling:

Motor fouling is expected to thermally lead to either or both increased winding losses, increased differential pressure for a given mass flow or increased temperature difference. A motor fouling is expected to lead to either high losses or cooling issues.

For losses, evolution of motor efficiency can be monitored.

Variable	Unit	Graph plot	Variable type
Motor efficiency	**Adim**	**Motor (Motor efficiency)**	**Y**
Motor torque	**N.m**	**Motor (motor torque)**	**X1**
Swirl parameter	**adim**	**Motor (swirl parameter)**	**X2**
Section 1 inlet temperature	**°C**	**Compressor (section 1/ inlet temperature)**	**X3**

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For temperature, motor fouling will tend to increase thermal insulation between cooling media (regulated) and winding. A drift of winding temperature should be visible and at certain point, winding temperature should take over onto TCV standard control. If this is the case, return temperature should tend to drop (overcooling) and will be visible by tracking evolution of return line temperature.

Variable	Unit	Graph plot	Variable type
Winding temperature	°C	Motor (Stator instrumentation/temperature)	Y
Shaft Power	kW	VSDS system (Shaft Power)	X1
Motor torque	N.m	Motor (motor torque)	X2
Swirl parameter	Adim	Motor (swirl parameter)	X3
Variable	Unit	Graph plot	Variable type
Return line temperature	°C	Cooling loop standard (Input for UCP/Return line temperature)	Y
Shaft Power	kW	VSDS system (Power electronics/Shaft Power)	X1
Motor torque	N.m	Motor (torque)	X2
Swirl parameter	Adim	Motor (swirl parameter)	X3

In the above graph, secondary variables are compressor inlet temperature and inlet pressure.

Finally, evolution of coupling temperature can be monitored. It is expected that the coupling temperature will show same trend as Return line temperature.

Variable	Unit	Graph plot	Variable type
Coupling temperature	kW	Cooling loop standard (Input for UCP Coupling temperature)	Y
Swirl parameter	Adim	Motor (swirl parameter)	X1
Shaft Power	kW	X VSDS system (Shaft Power)	X2
Motor torque	N.m	Motor (motor torque)	X3

The coupling temperature is assumed to have more direct link with windage effect and should evolve with above parameter.

Combination of coherence, health monitoring and above correlation should allow to detect either instrument issue in TCV or motor measurement or detected progressive fouling.

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5.6 Control valve health monitoring

The control valve hardware is the critical component of this architecture and shall regulate the return temperature.

Health of valve itself can be monitored by tracking the following:

Bias (static capacity)

Variable	Unit	Graph plot	Variable type
TCV	**°C**	**Cooling loop standard (Input from UCP-Coupling temperature)**	**Y**

Considering that TCV temperature is regulated, a direct time dependence of this parameter can be monitored to check no bias of the valve.

Dynamic control capacity

The capability of the valve to dynamically regulate the flow can be tracked by checking the dynamic response in mass flow to a variation of point.

Mass flow 1 or mass flow 2 or motor differential pressure can be used to be plotted against departure to temperature set point.

In case the TCV is instrumented, the % of opening of valve can be plotted against set point.

Variable	Unit	Graph plot	Variable type
Cooling flow	**Kg/s**	**Cooling loop standard (Input for UCP/Coupling flow)**	**Y**
TCV-setpoint	**°C**	**Cooling loop standard (Input for UCP/Temperature for TCV);temperature set point**	**X**
Variable	**Unit**	**Graph plot**	**Variable type**
% opening	**Kg/s**	**Cooling loop standard (Input for UCP/TCV position)**	**Y**
TCV-setpoint	**°C**	**Cooling loop standard (Input for UCP/Temperature for TCV)**	**X**

In previous plot, the interesting event will mostly be transient event (power or speed variation) to demonstrate that valve adapts properly (with correct and repetitive dynamic) to TCV variation. Such dynamic events like start-up or shut down or rapid speed change may differ significantly from time to time. It can turn to be difficult to monitor the valve dynamic response using only variable load case. A test scenario for vane could be created to force the dynamic and acquire a regular online signature.

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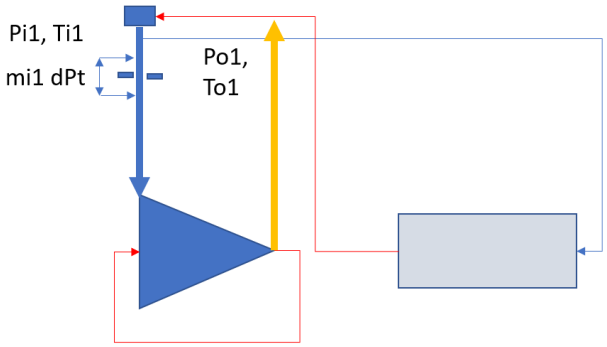
Ideally, an online testing sequence could be forced by imposing a forced setpoint variation (some degrees) at part load point and checking for response of system).

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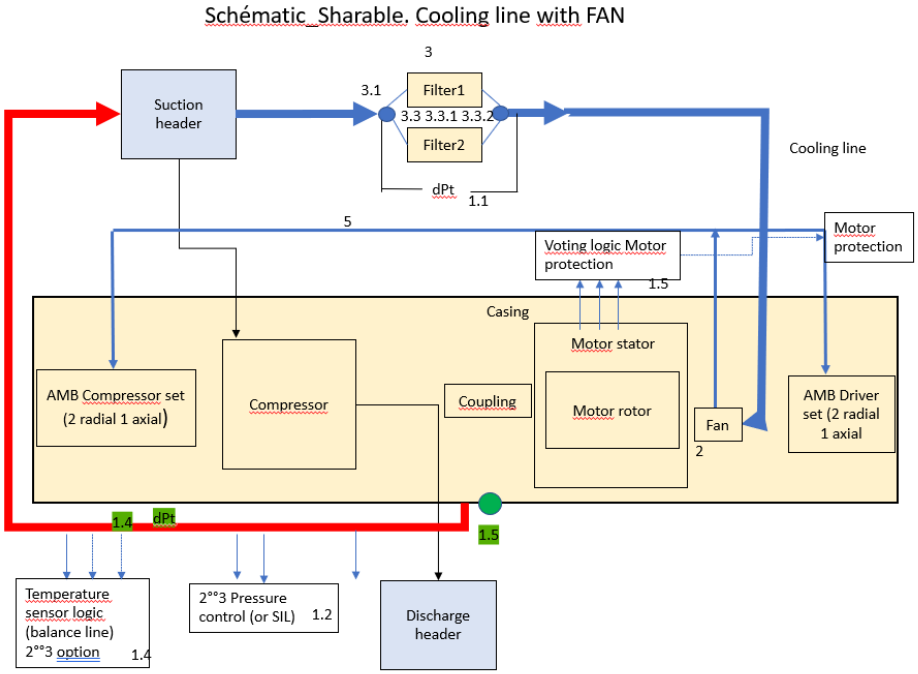
6 Cooling loop with Fan:

Cooling loop with Fan is a simpler configuration with respect to TCV option. It has nor flow regulation device (no TCV). On AMB side, Motor is equipped with a thrust bearing to allow to correctly localize the Fan rotor stator alignment. The motor thrust is not instrumented. Fan configuration is associated normally with an external instrument configuration.

External cooling scheme
(typical Fan cooling)



Heat balance: Motor Power-Gas Power= Coolloss



Rep		Description
Input for UCP		
	1	Filter differential pressure
	2	Pressure measurement 2°°3 voting
	3	Temperature (single or 2°°3 voted)
	4	Differential pressure return line(optional)
	5	Coooling temperature measurement
	6	Cooling mass flow 1
	7	Cooling mass flow 2
	8	Cooling power 1
	9	Cooling power 2
Fan		
	1	Fan flow coefficient

Figure 10: Fan instrument configuration

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6.1 Critical instrumentation and system

Motor by itself is a critical system whose performance need to be monitored. Contrary to the TCV option internal cooling scheme in which compressor section 1 performance is also depicting a image of motor performance, for Fan option an independent assessment is required.

Critical instruments present in the system are:

- Motor instrument
- Return line temperature instrument.
- Return line pressure instrument.

6.2 Non-critical instruments:

Casing coupling temperature monitoring is present as well as an optional measurement for return line differential pressure. This station is optionally instrumented.

6.3 List of calculated variables:

In this configuration of cooling, the control valve is suppressed, and a Fan is added. Fan is a relatively simple element that is more robust but can be sensitive to fouling. Calculation of fan flow coefficient is used to monitor performance and eventual fan fouling.

The torque of motor is used to monitor efficiency

Variable	Unit	Stored in system Cooling system Fan cooling	Variable type
Fan flow coefficient	**adim**	**Cooling loop Fan (mcool1 or m cool2)** **Antisurge (Section 1/ inlet pressure)** **Compressor (Section 1/ inlet temperature)** **ICL AMB (speed sensor)**	**Y**
Shaft Torque	**N.m or adim**	**VSDS (Power electronics)/AMBMotor(speed sensor)**	**X**

6.4 Coherence and health monitoring check

Health monitoring for 'dilution pressure':

A specific pressure measurement is present on cooling loop to ensure that pressure is not going below pre-defined pressure use to maintain pressurisation status (avoid air inlet).

Variable	Unit	Graph plot	Variable type
Return line pressure	**Pa**	**Cooling loop standard (Input for UCP /Return line Pressure)**	**Y**

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Section 1 inlet pressure	**Pa**	**Antisurge (Inlet Pressure)**	**X**

Health monitoring for Return Line temperature:

Health monitoring for temperature will be done simply by comparing casing temperature with return vane temperature. Casing and coupling temperature are assumed to follow one another

Variable	Unit	Graph plot	Variable type
TCV or votes temperature	**°C**	**Cooling loop standard (Input for UCP /Return line temperature))**	**Y**
Coupling temperature monitoring	**°C**	**Cooling loop standard (Input for UCP /Coupling temperature)**	**X1**

Health monitoring for motor temperature:

Motor temperature is expected to be more closely following the coupling and return line temperature

Variable	Unit	Graph plot	Variable type
Motor temperature	**°C**	**Motor(stator instrumentation/temperature)**	**Y**
Coupling temperature monitoring	**°C**	**Cooling loop standard (Input for UCP /Coupling temperature)**	**X1**
TCV control temperature	**°C**	**Cooling loop standard (Input for UCP / Return line temperature)**	**X2**

Cooling power and temperature:

In case the cooling flow is directly available by measurement (Mcool1) and the associated Cooling power, Mcool1 Pcool1 will be traced against respectively Mcool2 and Pcool2 obtained through energy balance.

6.5 Monitoring rules

For homogeneity with notation used before, the flange inlet condition will be this time noted pi' and ti' (after mixing).

Power balance:

For external cooling configuration Cooling power, the power balance allows to define Cooling power 2 by

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Cooling Power2=ShaftPower-Gas Power

At first step calculation of temperature prior to mixing (not measured) is performed. After this simple enthalpy mixing after cooling flow injection is performed

$$m.\big(H(pi',Ti') - H(Pi',Ti)\big) = CoolingPower2$$

$$m.H(pi',Ti') = mcool(H(Pi',TCV) + (m - mcool)H(Pi',Ti)$$

$$\frac{CoolingPower2}{(H(Pi,TCV) - H(pi'Ti)'} = mcool2$$

In case a measurement of cooling flow is available through differential pressure monitoring, it will give directly mcool1. Mcool1 allows to recalculate Cooling Power 1

Motor Fan performance:

The motor cooling loop FAN is not regulated, and the cooling circuit is passive (i.e circuit resistivity is proportional to square volume flow with a speed increase Fan head proportional to square speed and flow proportional to speed), it is expected that the flow coefficient of operation of the fan will be almost 'constant' or varies slightly with U/a0.

Variable	Unit	Graph	Variable type
Fan flow coefficient	**Adim**	**Cooling loop Fan (Input for UCP /Fan flow coefficient)**	**Y**

In case of a constant drift of flow coefficient towards low, flow, a fouling issue could be suspected. This fouling issue should be also demonstrated in temperature monitoring plot by a temperature increase of all temperature.

Motor Cooling:

In case of fouling issue that will affect either Fan capability or heat thermal exchange coefficient, it is expected that at minimum motor winding temperature will tend to increase when plot against main X. Compared to the configuration with TCV, to be noted that compressor inlet temperature is a new parameter which has direct influence onto winding temperature.

Analysis of Fan flow coefficient combined with motor temperature tracking can therefore fully diagnose.

Variable	Unit	Graph plot	Variable type
Coupling temperature	**kW**	**Cooling loop Fan (Coupling temperature)**	**Y**
Swirl parameter	**Adim**	**Cooling loop Fan (swirl parameter)**	**X1**
Shaft Torque	**N.m**	**VSDS system (Shaft Power)**	**X2**
Inlet motor temperature	**°C**	**Compressor (Section 1/Inlet temperature)**	**X3**

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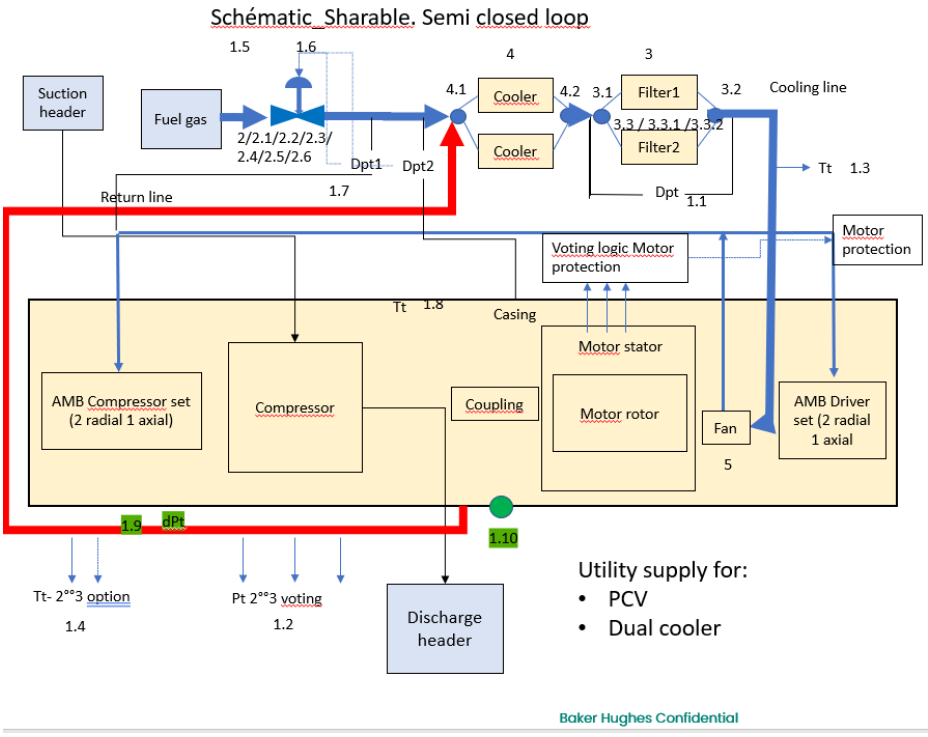
Variable	Unit	Graph plot	Variable type
Return line temperature	kW	Cooling loop Fan (Input for UCP /Coupling temperature)	Y
Swirl parameter	Adim	Motor (swirl parameter)	X1
Shaft Torque	N.m or adim	VSDS (Power electronics/Shaft Power)	X2
Inlet motor temperature	°C	Compressor (Section1/Inlet temperature)	X3

Variable	Unit	Graph plot	Variable type
Winding temperature	kW	Cooling loop Fan (Input for UCP /Coupling temperature)	Y
Swirl parameter	Adim	Motor(Swirl parameter)	X1
Shaft Torque	N.m or adim	VSDS system (Power electronics/Shaft Power)	X2
Inlet motor temperature	°C	Compressor (Section 1/Inlet temperature)	X3

7 Cooling loop with semi-closed loop:

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7.1 List of critical system and instrumentation:



Rep	Description
1	Input for UCP
1	Filter differential pressure
2	Pressure measurement 2°°3 voting
3	Temperature probe
4	Temperature (single or 2°°3 voted)
5	Fuel gas over-pressure
6	Differential pressure AMB
7	Differential pressure motor
8	Casing temperature measurement
9	Differential pressure (optional)
10	Cooling mass flow 1
11	Cooling mass flow 2
12	Cooling mass flow 3
13	Cooling power 1
14	Cooling power 2
15	Cooling power 3
2	Pressure control valve
1	Position transmitter
3	Fan
1	Fan Flow coefficient
4	Cooler (stored in process
1	Cooler circuit pressure
2	Cooler circuit water flow
3	Cooling water inlet temperature
4	Cooling water outlet temperature
5	Primary cool source temperature (atmosphere or water circuit)

Figure 11: Semi-closed loop critical system and instrumentation. Note that PCV position transmitter is not necessarily critical instrumentation.

The semi-closed loop is a derivation of the Fan option. In this configuration, the interaction between the compressor and cooling circuit is further reduced by the presence of cooler and a recycle mode on motor.

Influence of hot cooling gas on compressor is limited to the residual leakage existing between motor/AMB and compressor zone.

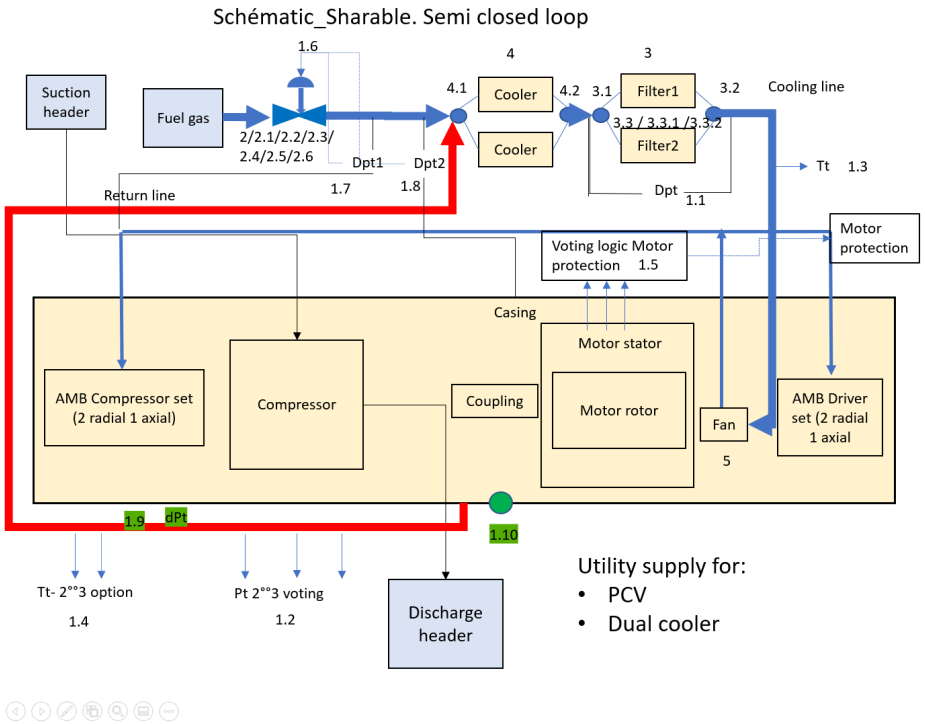
This leakage is controlled indirectly through the targeted differential pressure between bearing and compressor balance line.

Compared to a FAN version, the following system and instruments are present:

- PCV to control the external cooling gas
- Two differential pressure measurement uses to control PCV over-pressure
- A cooler system (single or dual cooler). The supply of the cooler may not be under BH responsibility. Most typical cooler would be air-cooler but water cooler can be also possible

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7.2 List of instrumentation:



Rep		Description
1		Input for UCP
	1	Filter differential pressure
	2	Pressure measurement 2°°3 voting
	3	Temperature probe
	4	Temperature (single or 2°°3 voted)
	5	Fuel gas over-pressure
	6	Differential pressure AMB
	7	Differential pressure motor
	8	Optional differential pressure
	9	Casing temperature measurement
	10	Cooling mass flow 1
	11	Cooling mass flow 2
	12	Cooling mass flow 3
	13	Cooling power 1
	14	Cooling power 2
	15	Cooling power 3
2		Pressure control valve
	1	Position transmitter
3		Fan
	1	Fan Flow coefficient
4		Cooler (stored in process
	1	Cooler circuit pressure
	2	Cooler circuit water flow
	3	Cooling water inlet temperature
	4	Cooling water outlet temperature
	5	Primary cool source temperature

Figure 12: instrumentation suivant la liste IPSB

7.3 Power balance:

For this configuration, compressor and motor performances are almost separated. All compressor performances will be followed using standard compressor parameters with no motor influence.

The power balance is identical to FAN configuration

$$Coolpower2 = ShaftPower - Gas\ Power$$

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It is possible from this CoolPower2 to calculate cooling flow by enthalpy equation:

$$Cooling\ flow2 = \frac{CoolPower2}{H(PTcref, Ttrcef) - H(Ptcin, Ttcin)}$$

Considering Fan very low head, it will be considered that PtCref=Ptcin in above equation and the pressure located in discharge pipe will be used.

In case a direct flow measurement is present on return line cooling flow 1 is measured directly and CoolPower1 is calculated by same equation:

$$Cooling\ flow1.\big(H(PTcref, Ttrcef) - H(Ptcin, Ttcin)\big) = CoolPower1$$

Finally in case data are available from the cooler, it is possible to measure Cooling power by water difference and get CoolPower3 and associated cooling flow 3

Name of calculated variable	Unit	System: Cooling loop Fan Input variables used for calculation of a calculated variable	Variable type
Cooling flow 1	Kg/s	Cooling loop semiclosedloop (Input for UCP /(Return line differential pressure, return line pressure, return line temperature))	Y or X
Cooling flow 2	Kg/s	Compressor (Gas Power) VSDS (Power electronics/shaft Power) Cooling loop semiclosedloop (Input for UCP /(Return line pressure, return line temperature))	Y or X
Cooling flow 3	Kg/s	Cooling loop semiclosedloop (Input for UCP /(cooler flow, cooler inlet temperature cooler outlet temperature))	
Coolpower 1	kW	Cooling loop semiclosedloop (Input for UCP /((cooling flow 1, Return line temperature, Return line pressure)) Compressor (Section 1/inlet temperature, Section1/mass flow)	X or Y
Coolpower 2	kW	Cooling loop semiclosedloop (Input for UCP /(cooling flow 2, Temperature for TCV)) Antisurge (Pinlet section 1) Compressor (Section1/Inlet temperature, mass flow)	X or Y
Coolpower 3	kW	Cooling loop semiclosedloop (Input for UCP /(cooling flow 2, Temperature for TCV)) Antisurge (section 1/Pinlet) Compressor (Section1/Inlet temperature, Section 1/mass flow)	X
Fan flow coefficient	Adim	Cooling loop semi closed loop (Input for UCP /(mcool1 or m cool2) Antisurge (Section 1 /inlet pressure) Compressor (Section 1/ inlet temperature)	X

Output parameters common to all cooling loop will be stored under Motor system.

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7.4 Coherence and health monitoring check

Health monitoring for 'dilution pressure':

A specific pressure measurement is present on cooling loop to ensure that pressure is not going below pre-defined pressure use to maintain pressurisation status (avoid air inlet).

Variable	Unit	Graph plot	Variable type
Return line pressure	Pa	Cooling loop semiclosedloop (Input for UCP /(Return line Pressure))	Y
Section 1 inlet pressure	Pa	Anti surge (Section1/(Inlet Pressure))	X

Health monitoring for Return Line temperature:

health monitoring for temperature will be done simply by comparing casing temperature with return vane temperature. Casing and coupling temperature are assumed to follow one another

Variable	Unit	Graph plot	Variable type
Return line temperature	°C	Cooling loop semiclosedloop (Input for UCP /(return line temperature))	Y
Coupling temperature monitoring	°C	Cooling loop semiclosedloop (Coupling temperature)	X

Health monitoring for motor temperature:

Motor temperature is expected to be more closely following the coupling and return line temperature.

Variable	Unit	Graph plot	Variable type
Motor temperature voted	°C	Motor (stator instrumentation/temperature probes) Nota: Either the highest or all temperature probes will be tracked	Y
Coupling temperature monitoring	°C	Cooling loop semi closed loop (Input for UCP /(Coupling temperature))	X1
Return line temperature	°C	Cooling loop semi closed loop (Input for UCP /(return line temperature)	X2

Differential pressure probes:

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Variable	Unit	Graph plot	Variable type
Differential Pressure DE	Pa	Cooling loop semi closed loop (Input for UCP /(Differential pressure DE))	Y
Differential Pressure NDE	Pa	Cooling loop semi closed loop (Input for UCP /(Differential pressure NDE))	X

Cooling Power:

Depending on available elements, the cooling power 1 2 and 3 can be calculated

Variable	Unit	Graph plot	Variable type
Cooling Power1	kW	Cooling loop semi closed loop (Input for UCP /(Cooling Power 1))	Y
Cooling Power2	kW	Cooling loop semi closed loop (Input for UCP /(Cooling Power 2))	Y
Cooling Power3	kW	Cooling loop semi closed loop (Input for UCP /(Cooling Power 3)	Y

7.5 Monitoring rules

Rules to follow motor fouling are like the one for Fan configuration.

Motor Fan performance:

The motor cooling loop FAN is not regulated, and the cooling circuit is passive (i.e circuit resistivity is proportional to square volume flow with a speed increase Fan head proportional to square speed and flow proportional to speed), it is expected that the flow coefficient of operation of the fan will be almost 'constant' or varies slightly with U/a0

Variable	Unit	Graph plot	Variable type
Fan flow coefficient	adim	Cooling loop semi closed loop (Fan flow coefficient)	Y

In case of a constant drift of flow coefficient towards low, flow, a fouling issue could be suspected. This fouling issue should be also demonstrated in temperature monitoring plot by a temperature increase of all temperature.

Motor Cooling:

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In case of fouling issue that will affect either Fan capability or heat thermal exchange coefficient, it is expected that at minimum motor winding temperature will tend to increase when plot against main X. Compared to the configuration with TCV, to be noted that compressor inlet temperature is a new parameter which has direct influence onto winding temperature.

Analysis of Fan flow coefficient combined with motor temperature tracking can therefore fully diagnose.

Variable	Unit	Graph plot	Variable type
Coupling temperature	kW	Cooling loop semi closed loop (Input for UCP /(Coupling temperature))	Y
Swirl parameter	Adim	Motor (swirl parameter)	X1
Motor torque	N.m or adim	VSDS system (Power electronics/Shaft Power)	X2
Inlet motor temperature	°C	Cooling loop semi closed loop (Input for UCP /(Coupling temperature))	X3

Variable	Unit	Graph plot	Variable type
Return line temperature	kW	Cooling loop semi closed loop (Input for UCP /(Coupling temperature))	Y
Swirl parameter	Adim	Motor (swirl parameter)	X1
Motor torque	N.m or adim	VSDS system (Power electronics/Shaft Power)	X2
Inlet motor temperature	°C	Cooling loop semi closed (Input for UCP /(Inlet temperature))	X3

Variable	Unit	Graph plot	Variable type
Winding temperature	kW	Cooling loop semi closed (Input for UCP /(Coupling temperature)	Y
Swirl parameter	Adim	Cooling loop semi closed (stator instrumentation/Swirl parameter)	X1
Motor torque	N.m or adim	VSDS system (Power electronicsShaft Power)	X2
Inlet motor temperature	°C	Cooling loop semi closed (Input for UCP Inlet temperature)	X3

In all above table, the line rotating speed and inlet pressure are likely contributing parameter considering that cooling mass flow is not regulated but passively generated by Fan. Considering the high number of variables impacting temperature, it is likely that a dedicated analytical model shall be used for tracking

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PCV:

PCV will be firstly monitor through a time analysis of the differential set point together with time plot:

Variable	Unit	Graph plot	Variable type
PCV position	%	Cooling loop semi closed (pressure control valve/position transmitter)	Y
Differential pressure DE	Pa	Cooling loop semi closed (Input for UCP /differential pressure DE)	Y
Differential pressure DE	Pa	Cooling loop semi closed (Input for UCP/ Differential pressure NDE)	Y
Inlet pressure s1	Pa	Antisurge (Section1/inlet Pressure)	Y
Buffer pressure	Pa	Process (External buffer network pressure)	Y

Previous plots are sufficient to judge of stability of valve in close to equilibrium position. However, stability of valve will be specifically verified during the strong transient event such as start-up or shut down for which full opening or closure shall be observed.

Capability to correctly control pressure also depends on available buffer network pressure compared to inlet pressure.

The use of analytical model might be necessary to properly track valve operation.

It is also likely that the valve dynamic and comparison of valve operation can be checked during the pressurization (signature of pressurization)

Coolers:

Data relative to cooler instruments are supposed to be in process as they are most of time customer supplied.

In case of availability of cooler data, a generic follow-up of a cooling loop is done. Details on details of the regulation and hardware present on cooling system to be adapted case by case:

Variable	Unit	Graph plot	Variable type
Pressure	Pa	Process (cooler loop pressure)	Y
Thermal power	Pa	Cooling loop semi closed (Input for UCP /Powercool 3)	Y
Cooling flow	Kg/s	Process (Cooler Cooling flow)	Y

The Pressure graph is most relevant to detect and eventual issue of leakage in the cooling system.

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Pressure as a function of flow can be traced to indicate a drift in pump performance in the circuit (i.e centrifugal pump, fixed speed)

Variable	Unit	Graph plot	Variable type
Pressure	**Pa**	**Process (cooling loop pressure)**	**Y**
Cooling flow	**Kg/s**	**Process (Cooler Cooling flow)**	**X**

Stability in time of inlet motor temperature is an objective of the coolerregulation.

As such this temperature will also be traced in time and during large power swing as function of Shaft power and/or Cooling power.

Variable	Unit	Graph plot	Variable type
Temperature	**°C**	**Cooling loop semi closed (cooling temperature)**	**Y**
Shaft power	**kW**	**VSDS (Power electronics/shaft Power)**	**Y**
Temperature	**°C**	**Process (external temperature)**	**X**

The last graph will be of specific interest to verify repeatability of temperature control (prime control target) in time.

The external temperature (primary source of cooling) is at base variation for temperature achievable downstream.

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8 Sampling rate requirement for Historian recording:

Performance data are used only for stable situation and do not require high sampling rate. However, some of data are also used for other system monitoring with higher dynamic requirement. Sampling time can be relaxed for temperature probes only used for performance monitoring.

- Compressor inlet Volume flow, inlet, or outlet temperature: Those data are used for dynamic monitoring of anti-surge system: sampling to UCP cycle time (1 per second).
- Compressor rotating speed: sampling to UCP cycle time (1 per second).
- Motor winding temperature, gas return line temperature (TCV temperature objective): Sampling 1 per 10 second to one per minute max (use to detect discordance of TCV valve)
- Process gas temperatures: Sampling 1 per minute to 1 per 15-minute max.

For general performance post-treatment, in case sampling rate exceeds requirement, post-treatment will be done using time average value for stable period using at minimum a 15 minutes time average.

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Annex I PSB and IPSB standard high-level structure:

Refer to [R1] for excel version of documentation for all level for PSB and IPSB to be used for reliability and monitoring of ICL.

System Id	System description	Level 2 id	Description
I	Compressor	1	Main flange
		2	Compressor Cover
		3	AMB IGV Support Optional for LPR
		4	Discharge support
		5	Diaphragm pack Optional for LPR
		6	Rotor
		7	Coupling trap Optional for LPR
		8	High-speed coupling Optional for LPR
		9	Inlet/outlet/balance line
		10	Compressor support
II	ICL-Compressor-AMBs (optional for LPR)		
		1	Radial-AMB-DE
		2	Radial-AMB-NDE
		3	Axial-AMB
		4	Connectors (AMB power)
		4	Connectors (AMB instrumentation)
III	Motor		
		1	Rotor
		2	Motor Casing
		3	Stator Instrumentation
		4	Motor casing cover
		5	Air Gap
		6	Motor legs
		7	Drains
		8	Connectors (stator temperature)

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IV	Motors-AMBs		
		1	Radial-AMB-DE
		2	Radial-AMB-NDE
		3	Axial-AMB (Optional)
		4	Connectors (AMB power)
		4	Connectors (AMB instrumentation)
V	Junction Boxes		
		1	Instrument Junction Box
		2	AMB power Junction Boxes
		3	Gas detectors
		4	HV Junction box
VI	Anti-surge		
VII	Cooling loop standard	1	Input for UCP
		2	Temperature control valve
		3	Dual Filter
		4	Cooling distribution pipes
VII bis	Cooling loop with Fan	1	Input for UCP
		2	Fan
		3	Dual or single Filter
		4	Cooling distribution pipes
VII ter	Cooling loop-Semi-closed loop		
		1	Input for UCP
		2	Pressure control valve
		3	Dual Filter
		4	Dual Cooler
		5	Fan
		6	Cooling distribution pipes

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VIII	AMB Compressor Cabinet	1	Control Unit
		2	Power amplifiers
		3	Interface
		4	I/O interface Wago
		5	Automate MBSCI type
		6	Internal power supply . Includes cooling and fans
		7	Batteries and charger
IX	AMB Motor Cabinet	1	Control Unit
		2	Power amplifiers
		3	Interface
		4	I/O interface Wago
		5	Automate MBSCI type
		6	Internal power supply . Includes cooling and fans
		7	Batteries and charger
X	VSDS	1	Internal cooling loop, de-ionised
		2	External cooler system (air or Water)
		3	Power electronics (include IGBT)
		4	Control Unit (including Pibe)
		5	Filter (sin or dv/dt)
		6	Main Transformer
XI	UCP and other electric or electronic systems	1	Power supply and transformers
		2	PLC Simplex or Duplex
		3	I/O numerical and analogical
		4	PLC safety
		5	Cabinet cooling system
		6	Low Voltage switch gear (often wrongly called MCC) for distibution protection and contactor
		7	UPS ininterupted power supply
XII	Process / Customer	1	Process instruments and data

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